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Distribution-Free Inventory Risk Pooling in a Multilocation Newsvendor

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Abstract. We study a multilocation newsvendor network when the only information available on the joint distribution of demands are the values of its mean vector and covariance matrix. We adopt a distributionally robust model to find inventory levels that minimize the worst-case expected cost among the distributions consistent with this information. This problem is NP-hard. We find a closed-form tight bound on the expected cost when there are only two locations. This bound is tight under a family of joint demand distributions with six support points. For the general case, we develop a computationally tractable upper bound on the worst-case expected cost if the costs of fulfilling demands have a nested structure. This upper bound is the optimal value of a semidefinite program whose dimensions are polynomial in the number of locations. We propose an algorithm that can approximate general fulfillment cost structures by nested structures, yielding a computationally tractable heuristic for distributionally robust inventory optimization on general newsvendor networks. We conduct experiments on networks resembling U.S. e-commerce distribution networks to show the value of a distributionally robust approach over a stochastic approach that assumes an incorrect demand distribution.

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1. Introduction

Consider a firm deciding the inventory levels in a network of inventory nodes before demand is realized in multiple customer locations. The demand in one customer location can be fulfilled by any inventory node. After demand is realized, the firm needs to decide how to fulfill the demands using the inventory levels. The firm incurs overage costs, underage costs, and fulfillment costs.

One can think of several applications for this problem. For example, in a make-to-order system, the firm is deciding the capacities at plants to meet the demand of multiple products, and after demands are realized, the firm decides the production quantities of products in each plant. In a retail network, the seller decides the inventory levels in multiple nodes of a product that does not have a chance to be replenished during the selling season. This is true of products whose selling season is much shorter than the lead time for replenishment (e.g., fast fashion products or other seasonal products). After demands are realized in multiple locations, the seller decides either transshipment quantities among a network of stores or, in the case of e-commerce, total shipments from fulfillment centers to customer locations. In all these settings, a common characteristic is that there is *inventory risk pooling*, which is the motivation behind strategies such as transshipment, process flexibility, component commonality, and drop shipping.

If the firm knows the joint probability distribution of the vector of demands, then the optimal inventory levels can be solved via the following two-stage stochastic program:

$$\min_{\mathbf{y}\geq 0} \ \mathbb{E}_f\big[C(\mathbf{y},\tilde{\mathbf{D}})\big], \tag{1}$$

where $\mathbb{E}_f$ is the expectation operator under the joint probability distribution f, $\mathbf{y} = (y_i)_{i=1}^m$ is the vector of inventory levels in m nodes, $\tilde{\mathbf{D}} = (\tilde{d}_j)_{j=1}^n$ is the vector of random demands in n locations, and for demand

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realization $\mathbf{D} = (d_j)$, $C(\mathbf{y}, \mathbf{D})$ is the optimal second-stage cost defined as

$$\begin{aligned} C(\mathbf{y}, \mathbf{D}) := \min_{\mathbf{Z} \geq 0} \quad & h \cdot \sum_{i=1}^{m} \left(y_i - \sum_{j=1}^{n} z_{ij} \right) \\ & + p \cdot \sum_{j=1}^{n} \left(d_j - \sum_{i=1}^{m} z_{ij} \right) \\ & + \sum_{i=1}^{m} \sum_{j=1}^{n} s_{ij} z_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^{n} z_{ij} \leq y_i, \quad \forall i = 1, \ldots, m, \\ & \sum_{i=1}^{m} z_{ij} \leq d_j, \quad \forall j = 1, \ldots, n. \end{aligned} \tag{2}$$

The objective function is the sum of the overage cost (where $h > 0$ is the per-unit overage cost), the penalty cost for unmet demand ($p > 0$ is the per-unit penalty cost), and the fulfillment cost ($s_{ij} > 0$ is the per-unit cost of fulfilling demand in location j using inventory in node i). The recourse decision z_{ij} is how many units to use from inventory node i to fulfill the demand in customer location j. We refer to (1) as the *multilocation newsvendor problem with inventory risk pooling*, since the optimal solution reflects pooling benefits due to shared inventory among correlated locations.

In reality, firms do not know the demand distribution, and at best, they may only have partial information about the distribution such as its mean and covariance. A common heuristic is to assume that the random demand vector has a multivariate normal distribution. However, the normal distribution assumption can lead to solutions that overestimate pooling benefits if the true demand distribution is nonnormal. Eppen (1979) showed that if the distribution is light-tailed (including the normal distribution), pooling can lead to savings in expected cost that scale with $\sqrt{n}$, where n is the number of random demands being pooled. However, there are studies that show evidence of real-life demands exhibiting nonnormal distributions in a variety of industries (Agrawal and Smith 1996, Bimpikis and Markakis 2015, Natarajan et al. 2017). Bimpikis and Markakis (2015) show that the pooling benefits can scale significantly lower than $\sqrt{n}$ when the demand distribution is heavy-tailed. Hence, assuming a specific distribution for the random demand can lead to decisions that may be suboptimal under the true (unknown) demand distribution.

If the firm only had information about the mean and covariance, how should it decide on inventory levels that reflect the benefit of inventory risk pooling? To address this question, we adopt a minmax distributionally robust approach that aims to choose inventory levels that minimize the maximal expected cost over all demand distributions consistent with the information known to the firm. Mathematically, this is done by solving

$$C^* := \min_{\mathbf{y} \geq 0} \sup_{f \in \mathcal{F}_{\geq 0}} \mathbb{E}_f[C(\mathbf{y}, \tilde{\mathbf{D}})], \tag{3}$$

where $\mathcal{F}_{\geq 0}$ is the set of all distributions with nonnegative support, mean $\mathbf{m}$, and covariance $\mathbf{\Sigma}$:

$$\begin{aligned} \mathcal{F}_{\geq 0} := \{ f : \mathfrak{R}^n_{\geq 0} \mapsto \mathfrak{R}_{\geq 0} \mid \mathbb{E}_f(1) = 1,\ \mathbb{E}_f(\tilde{\mathbf{D}}) = \mathbf{m}, \\ \mathbb{E}_f(\tilde{\mathbf{D}}\tilde{\mathbf{D}}^\top) = \textstyle\sum + \mathbf{m}\mathbf{m}^\top \}. \end{aligned} \tag{4}$$

We denote the optimal value of (3) as C^* and its optimal solution as $\mathbf{y}^*$. If the firm chooses the initial inventory level $\mathbf{y}^*$, then it can be guaranteed that the expected cost is no larger than C^* under any joint demand distribution with mean-covariance $(\mathbf{m}, \mathbf{\Sigma})$. Since the inventory levels are chosen to be robust to any specific distribution, we refer to (3) as the *distributionally robust* multilocation newsvendor problem with inventory pooling.

Our goal in this paper is, using a distributionally robust framework, to gain new insights into risk pooling when the distribution is ambiguous. Although many past works have adopted the distributionally robust approach in a newsvendor context (e.g. Scarf 1958, Gallego and Moon 1993, Bertsimas and Thiele 2006, Hanasusanto et al. 2015, Natarajan et al. 2017), to the best of our knowledge, none have used it to study inventory risk pooling. The challenge is that there are multiple demands which interact, not only because they are correlated, but also because of the shared inventory. Hence, even for small networks, the problem is difficult to analyze and solve. In fact, the distributionally robust problem (3) is NP-hard in general, so the best that one can hope for are computationally tractable heuristics for general networks.

1.1. Main Results and Contributions

1. Scarf (1958) develops a closed-form expression for the worst-case expected newsvendor cost with a single location ($n = 1$) given mean and variance information. However, since it is for a single location, it does not reveal how pooling the inventory of correlated locations impacts the worst-case expected cost. In Section 4 we study the robust problem with two locations ($n = 2$) that share inventory. We develop a closed-form expression for the worst-case expected cost that relies only on the means, variances, and *correlation* of the two demands.

We show that the bivariate demand distribution that incurs the highest expected cost has at most six support points. This allows us to study how the cost parameters and the inventory levels affect the worst-case distribution. Hence, we gain new insights into how demand ambiguity affects the optimal inventory

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solution under risk pooling. In particular, we show that the optimal solution is to keep the same safety stock in the locations so that $y_1^* - m_1 = y_2^* - m_2$.

2. For the general model, we focus our analysis on the class of nested fulfillment cost structures under which the second-stage cost (2) can be written as the sum of piecewise linear terms. In Section 2.1, we develop a simple algorithm (based on hierarchical agglomerative clustering) to approximate any general distance-based fulfillment cost structure as a nested cost structure. We show empirically that this structure tightly approximates the expected total fulfillment cost under a variety of distributions. A nested fulfillment cost structure not only yields tractability for our robust problem, but also for stochastic systems.

3. Even when restricted to the class of nested fulfillment cost structures, the distributionally robust multilocation newsendor problem is NP-hard. However, we reveal its connection to the worst-case expected value of a 0-1 linear program under objective uncertainty, which is equivalent to a completely positive program under mean and covariance information (Natarajan et al. 2011). This connection allows us to utilize techniques from Natarajan and Teo (2017) to develop a computationally tractable heuristic for our problem under nested fulfillment cost structures. The heuristic is computationally tractable since it relies on solving a semidefinite program (SDP) whose size is polynomial in the number of locations. Combined with the hierarchical agglomerative clustering technique, this yields a computationally tractable heuristic for general (not necessarily nested) cost structures. To the best of our knowledge, this is the first computationally tractable heuristic demonstrated for general multilocation newsvendor models with only information on mean, covariance, and nonnegative support.

4. We conduct numerical experiments to show that the distributionally robust heuristic can lead to significant savings in expected cost as compared with stochastic solutions that assume an incorrect distribution. The parameters of the experiment are derived from data of a U.S.-based online retailer's fulfillment center network, U.S. population data, and UPS shipping cost data. We demonstrate that if the firm only knows the mean and covariance, it can better capture the benefit of risk pooling through a distributionally robust approach instead of assuming a multivariate normal distribution, which is commonly done in practice.

1.2. Literature Review

Distributionally robust inventory models have a long history, beginning with Scarf (1958), who derives a closed-form expression for the robust single-item newsvendor solution with only mean and variance information. In a multi-item newsvendor setting, Gallego and Moon (1993) build on the results of Scarf (1958) by assuming that, for each item, the mean and variance of the demand is known, and there is a constraint on the total ordering costs. Given information on the correlations of demands, the distributionally robust multi-item newsvendor problem turns out to be much harder than the single-item problem. In fact, Hanasusanto et al. (2015) show that the problem is NP-hard, and they propose an approximation by using quadratic decision rules, which can be viewed as applying Scarf's bound for each item. Natarajan and Teo (2017) study the multi-item robust newsvendor assuming mean and covariance information is known. They develop semidefinite programming relaxations by showing a connection of the multi-item newsvendor problem with the Boolean quadric polytope and using semidefinite relaxations of the Boolean quadric polytope. Their approach provides tighter bounds than applying Scarf's bound to each item and captures dependence information. Although the bound is not tight in general, the size of the SDP scales polynomially in the number of items.

A robust framework is typically used to study multiperiod inventory systems, due to complexities arising from the dynamics of a distributionally robust setting. Bertsimas and Thiele (2006) assume an upper and lower bound on per-period demand and, inspired by Bertsimas and Sim (2004), a budget of uncertainty on the total absolute deviation of demand from nominal values. They show that the robust problem can be reformulated tractably as a linear program or, if there are fixed costs, as a mixed-integer linear program. Bienstock and Özbay (2008) generalize this approach by studying more general uncertainty sets and period-dependent costs. Mamani et al. (2016) derive closed-form solutions under a simpler setting (no fixed cost, single location) assuming a similar uncertainty set: upper and lower bounds on the per-period demand and bounds on the cumulative demand in each period. In order to achieve tractability, the aforementioned multiperiod robust inventory problems assume only knowledge of the demand distribution support. See and Sim (2010) assume knowledge of mean, covariance and forward/backward deviations of the random demand sequence, though the problem is intractable because of the multiperiod dynamics. They introduce a tractable heuristic by approximating the recourse decisions via linear decision rules (Ben-Tal et al. 2004, Chen et al. 2008). Their approach can be extended to a multiechelon network (i.e., inventory flows go in one direction), similar to the setting of Bertsimas and Thiele (2006). In our paper, we consider networks where inventory can flow between locations in either direction.

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There is less literature on robust inventory problems on a network where there are recourse network flows after demands are realized, which is the setting of our study. Chou et al. (2006) develop tractable heuristics for a multilocation multiperiod transshipment problem by assuming that the transshipment quantities are approximated via linear decision rules on primitive uncertainties with known support and forward/backward deviations. A contrast of our work to theirs is that we assume simpler information (i.e., mean and covariance of the demand vector), rather than information that is not intuitive to decision-makers (i.e., forward/backward deviations). Another important distinction is that Chou et al. (2006) use a similar approach as See and Sim (2010) to deal with the multiperiod dynamics, by restricting demands to depend linearly on the primitive uncertainties in combination with loose bounds on expected transshipment quantities. We do not impose such a restriction in our paper. Mak and Shen (2014) study pooling in a multiple-location inventory system where both demand and yield loss are random. Using a distributionally robust setting (knowing only moments of the marginal demand distributions and information on dependence), they show that the expected cost is increasing in the degree of demand (or yield) dependence. The focus, however, of their work is on establishing a directional relationship between the expected cost and demand dependence, whereas in our work, our goal is to find a closed-form expression or tractable heuristics for solving the robust multilocation newsvendor problem.

Our study is also related to the literature on inventory pooling. Eppen (1979) shows that if n is normal and uncorrelated demands are pooled, the benefit from inventory pooling is $\sqrt{n}$, and the benefit decreases with increasing positive correlation among demands. Corbett and Rajaram (2006) extend Eppen's result to more general distributions. Yang and Schrage (2009) study various cases of "inventory anomaly" (a situation where pooling leads to an increase in inventory as opposed to a reduction), one of which is for right-skewed demand distributions with product substitution. Berman et al. (2011) find through numerical simulations that the normal distribution misestimates the benefits of pooling stemming from a reduction in variance. Bimpikis and Markakis (2015) find that the benefit from pooling under heavy-tailed demand distributions can be significantly lower than $\sqrt{n}$. Specifically, they show that the benefit from pooling decreases as the tail of the demand becomes heavier. All these studies indicate that pooling benefits crucially depend on the distribution of the demands being pooled. We contribute to this literature by analyzing pooling benefits in a distributionally robust setting, which reveals new insights into risk pooling.

We also briefly mention the related literature on process flexibility, for which there is a long history. Recent works include Wang and Zhang (2015), Yan et al. (2018), and Shi et al. (2019). However, the goal in such works is to study the effectiveness of specific sparse network designs (e.g., a long chain network), whereas our focus is on deriving robust inventory placement strategies.

2. Nested Fulfillment Cost Structures

Our goal is to develop tractable heuristics for the NP-hard problem (3). However, it is difficult to analyze the problem and to develop heuristics without imposing additional structure to the model. In this section, we introduce a structure on the fulfillment costs that allows us to significantly simplify the cost function. Although this structure still results in an NP-hard problem (as discussed in Section 3), it aids us in deriving the computationally tractable heuristic in Section 5.

Consider a network with n inventory nodes and n customer locations. Specifically, we pair each customer location to exactly one inventory node and assign them the same index. A pairing means that the fulfillment cost between the pair is the smallest compared with fulfillment costs to other locations. By using our notation in (2), this means that $s_{ij}(= s_{ji}) \geq s_{jj}$ for all $i, j \in [n]$. (For any integer n, $[n]$ denotes the set $\{1, 2, \ldots, n\}$.) An example of this type of network is a geographical network where there is one inventory node located in each customer location and fulfillment costs are proportional to distances. Our analysis can be extended to general networks by modeling customer locations without an inventory node as having zero inventory.

In this section, we introduce the notion of a *nested fulfillment cost structure*. We explain this concept using the example of the following fulfillment cost matrix for $n = 4$ locations:

$$\mathbf{S} = \begin{bmatrix} 1 & 2 & 5 & 5 \\ 2 & 1 & 5 & 5 \\ 5 & 5 & 1 & 2 \\ 5 & 5 & 2 & 1 \end{bmatrix}. \tag{5}$$

We observe from this matrix that *in-location* fulfillment (i.e., demand fulfilled from the inventory node at the same location) has the lowest cost of 1. This is followed by a cost of 2, which is *cross-location* fulfillment between locations 1 and 2 and between locations 3 and 4. Fulfillment between any other pairs of locations incurs the highest cost of 5. Matrix $\mathbf{S}$ induces a nested hierarchy fulfillment, as illustrated in Figure 1. We refer to Figure 1 as a *three-level nested cost structure*.

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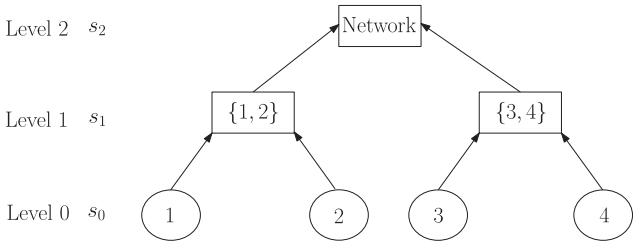
Figure 1. Example of a Three-Level Nested Fulfillment Cost Structure Induced by $\mathbf{S}$ in (5)



To read this figure, suppose that demands in the four locations are realized. Fulfillment will first take place in level 0 at a cost of $s_0 = 1$. Then, if there is remaining unmet demand, it is fulfilled in level 1 at a cost of $s_1 = 2$. For example, if location 1 does not have enough inventory to meet its own demand (level 0), then it is fulfilled by any remaining inventory in location 2 (level 1). If there is remaining unmet inventory even after level 1 fulfillment, it is fulfilled in level 2 at cost $s_2 = 5$.

We now describe a general L-level nested fulfillment hierarchy. For a given level ℓ, where $\ell = 0, 1, \ldots, L-1$, the locations are partitioned into n_ℓ sets, where $n_\ell \le n$. (In the previous example, there are four sets (i.e., $\{1\}, \{2\}, \{3\}, \{4\}$) in the level 0 partition and two sets (i.e., $\{1, 2\}, \{3, 4\}$) in the level 1 partition.) We denote the n_ℓ sets in partition ℓ as $\{\mathcal{I}_1^{(\ell)}, \mathcal{I}_2^{(\ell)}, \ldots, \mathcal{I}_{n_\ell}^{(\ell)}\}$. By definition, the sets in a partition must cover all locations, and the intersection of any two sets is empty.

The cost structure is nested because of the following property: any set in level ℓ is the union of sets in the preceding level $\ell - 1$. That is, $\mathcal{I}_k^{(\ell)} = \cup_{m \in \mathcal{K}_k^{(\ell)}} \mathcal{I}_m^{(\ell-1)}$, where $\mathcal{K}_k^{(\ell)}$ are the indices of the level $\ell - 1$ children of $\mathcal{I}_k^{(\ell)}$. Note that $n_0 > n_1 > \cdots > n_{L-1}$, since there are fewer sets in higher order levels. We assume that, for level 0, $n_0 = n$ and $\mathcal{I}_i^{(0)} = \{i\}$ for all $i \in [n]$. For the final level $L-1$, $n_{L-1} = 1$ and $\mathcal{I}_1^{(L-1)} = [n]$. Any L-level nested fulfillment cost structure can be represented as an L-level tree similar to Figure 1. (Another example is Figure 2(b).)

The nested cost structure can also be represented through the assignment matrices $\Xi = \{\mathbf{E}_0, \mathbf{E}_1, \ldots, \mathbf{E}_{L-1}\}$. We define the level ℓ assignment matrix $\mathbf{E}_\ell$ as the binary matrix of size $n_\ell \times n$, where the (k, i) entry is equal to 1 if and only if $i \in \mathcal{I}_k^{(\ell)}$. Note that $\mathbf{E}_0$ is the $n \times n$ identity matrix and that $\mathbf{E}_{L-1}$ is the row vector of all ones.

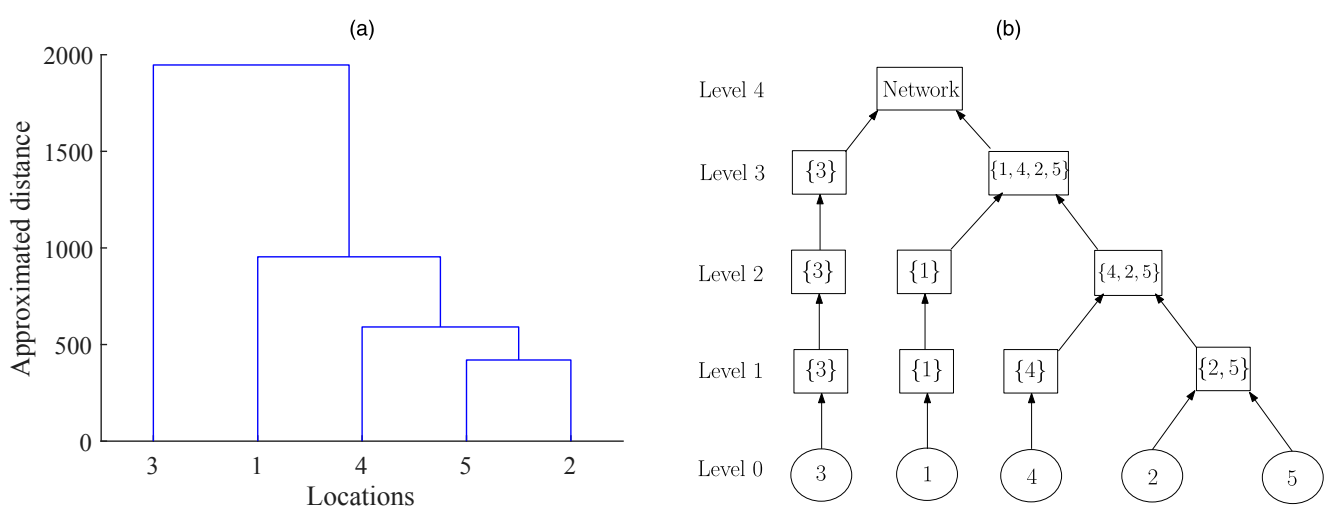
We next discuss the fulfillment costs. If two locations are in set $\mathcal{I}_k^{(\ell)}$, then the per-unit cost of fulfillment between the two locations is $s_{\ell,k}$. We denote $\mathbf{s} = \{s_{\ell,k}\}$ as the set of all fulfillment costs. In Figure 1, locations 1 and 2 are in the same level 1 set, so the fulfillment cost between them is s_1. Note that, in the example, all level 1 costs are equal; however, in general, we allow sets in the same level to have different costs. To induce the nested hierarchy, we need the following assumption.

Assumption 1. *Denote $k^{(\ell)}(i)$ as the level ℓ set index of location i. For any location i, we have $s_{0,k^{(0)}(i)} \le s_{1,k^{(1)}(i)} \le \cdots \le s_{L-1} \le p + h$.*

Assumption 1 states that, for demand originating from location i, the fulfillment cost as you go up the tree is increasing. Hence, this induces the nested hierarchy for fulfillment. We can assume $s_{L+1} \le p + h$ without loss of generality. This is because, if $s_{\ell,k} > p + h$ for some node (ℓ, k) in the tree, then the children nodes of set k will never use fulfillment in level ℓ or above, and the problem decomposes as each level $\ell - 1$ child of set k can be removed to form new networks.

The nested hierarchy, Ξ, and the fulfillment costs, $\mathbf{s}$, characterize the nested cost structure. As we discuss

Figure 2. (Color online) Approximating a Distance-Based Fulfillment Cost Structure with a Nested Cost Structure



Notes. Panel (a) is the dendrogram obtained by hierarchical clustering based on the distance matrix $\mathcal{R}$. Panel (b) is the corresponding five-level nested structure.

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next, this class of cost structures can approximate general fulfillment cost structures.

2.1. Approximating Distance-Based Fulfillment Costs with a Nested Structure

Suppose that the fulfillment cost is a function of the distance between two locations. That is, $s_{ij} = \phi(r_{ij})$ for any i, j, where r_{ij} is the distance between the locations and ϕ is an increasing function. We denote by $\mathcal{R} = (r_{ij})_{ij}$ the distance matrix and $\mathbf{S} = (s_{ij})_{ij}$ the fulfillment cost matrix. We next develop a simple algorithm that employs hierarchical agglomerative clustering (Johnson 1967) so that, for a general distance-based fulfillment cost structure, we can find approximate fulfillment costs that have an n-level nested structure. We empirically show that this nested structure tightly approximates the actual expected total fulfillment cost under a variety of distributions.

The algorithm can be found in Section EC.1 of the e-companion (Algorithm 1). It proceeds by progressively clustering the two closest nodes into a single node starting from the n leaf nodes corresponding to locations, until there remains only one cluster node which encompasses all the locations. In each step of the algorithm, the number of nodes is reduced by 1. Note that, in order to choose the two closest nodes in each step, we need a notion of distance between clusters of nodes. A variety of measures can be considered to define the distance between two clusters, namely, the minimum or maximum or average distance between the nodes in the two clusters, Ward's method, distance between the center of masses of the two clusters, and so on.

We demonstrate the algorithm using an example, where we define the distance between two clusters as the average distance measure, namely, the unweighted pair group method with arithmetic mean (UPGMA). The UPGMA distance between clusters $\mathcal{I}_1, \mathcal{I}_2$ is $\hat{r}_{\mathcal{I}_1,\mathcal{I}_2} = \frac{1}{|\mathcal{I}_1|} \cdot \frac{1}{|\mathcal{I}_2|} \cdot \sum_{i\in\mathcal{I}_1, j\in\mathcal{I}_2} r_{ij}$. In other words, it is the average distance between any two pairs of locations in $\mathcal{I}_1$ and $\mathcal{I}_2$.

Consider the following distance matrix among five nodes and the corresponding fulfillment cost matrix constructed by the equation $s_{ij} = 10 + 0.005 \cdot r_{ij}$:

$$\mathcal{R} = \begin{bmatrix} 0 & 1,220 & 1,411 & 770 & 872 \\ 1,220 & 0 & 2,404 & 624 & 420 \\ 1,411 & 2,404 & 0 & 1,785 & 2,187 \\ 770 & 624 & 1,785 & 0 & 557 \\ 872 & 420 & 2,187 & 557 & 0 \end{bmatrix},$$

$$\mathbf{S} = \begin{bmatrix} 10.0 & 16.1 & 17.1 & 13.8 & 14.4 \\ 16.1 & 10.0 & 22.0 & 13.1 & 12.1 \\ 17.1 & 22.0 & 10.0 & 18.9 & 20.9 \\ 13.8 & 13.1 & 18.9 & 10.0 & 12.8 \\ 14.4 & 12.1 & 20.9 & 12.8 & 10.0 \end{bmatrix}.$$

Applying Algorithm 1 with UPGMA as the distance metric, we obtain the dendrogram shown in Figure 2(a). The dendogram depicts the clustering in each step (level). The dendogram also depicts the distance between the entities being clustered: in level 2 for instance, $\{4\}$ is clustered with $\{5, 2\}$ at the UPGMA distance $\hat{r}_{4,\{5,2\}} = 590.6$. This means that any fulfillment between nodes 4 and 5 or between nodes 4 and 2 takes place at a cost $\hat{s}_{4,\{5,2\}} = 10 + 0.005 \cdot \hat{r}_{4,\{5,2\}}$. Thus, we have an approximation of the distance and fulfillment cost matrices:

$$\hat{\mathcal{R}} = \begin{bmatrix} 0 & 954 & 1,947 & 954 & 954 \\ 954 & 0 & 1,947 & 591 & 420 \\ 1,947 & 1,947 & 0 & 1,947 & 1,947 \\ 954 & 591 & 1,947 & 0 & 591 \\ 954 & 420 & 1,947 & 591 & 0 \end{bmatrix},$$

$$\hat{\mathbf{S}} = \begin{bmatrix} 10.0 & 14.8 & 19.7 & 14.8 & 14.8 \\ 14.8 & 10.0 & 19.7 & 13.0 & 12.1 \\ 19.7 & 19.7 & 10.0 & 19.7 & 19.7 \\ 14.8 & 13.0 & 19.7 & 10.0 & 13.0 \\ 14.8 & 12.1 & 19.7 & 13.0 & 10.0 \end{bmatrix}.$$

Algorithm 1 always results in a nested fulfillment cost structure with $L = n$ levels (see Figure 2(b)). We can generate a nested structure for any general L. We do this by cutting the dendrogram horizontally at $L - 2$ places across the y-axis: the ℓth line from the bottom gives rise to a partition of the set of locations by cutting through links that cluster these partitioned sets further. This partition gives us the clusters at level ℓ. We provide an example in the e-companion (Section EC.1.2).

The nested fulfillment cost structure is a good approximation whenever the geographical region inherently contains a hierarchical cluster structure. This is common in the United States, where population centers are dispersed, so intercluster distances are much higher than intracluster distances. Indeed, approximation errors arise when we ascribe a single fulfillment cost to fulfillment between any two locations in two different clusters. However, the quantity of unmet demand fulfilled in these higher levels is small when more demands are being pooled within each cluster, and as a result, the error in approximation of the total fulfillment cost is small. In Section 6.1, we show that the gap in the expected total fulfillment cost is less than 3% when $L = n$ for a general distance-based fulfillment cost on a U.S. inventory network, even when the sensitivity of the cost to distances is high.

3. Distributionally Robust Multilocation Model

Our restriction to nested fulfillment costs allows us to simplify the second-stage cost function (2). In particular,

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the cost function can be written in closed-form, as formalized in the following lemma.

Lemma 1. *Under the L-level nested fulfillment cost structure,*

$$C(\mathbf{y},\mathbf{D}) = h\cdot\mathbf{e}^\top(\mathbf{y}-\mathbf{D}) + \mathbf{s}_0^\top\mathbf{D} + \sum_{\ell=0}^{L-1}\eta_\ell^\top(\mathbf{E}_\ell\mathbf{D}-\mathbf{E}_\ell\mathbf{y})^+, \quad (6)$$

where $(\mathbf{x})^+ = (x_i^+)$ *for any vector* $\mathbf{x}$, $\eta_{L-1} = p + h - s_{L-1}$, *and for* $\ell \le L-2$, $\eta_\ell = (\eta_{\ell,k})_{k\in[n_\ell]}$ *with* $\eta_{\ell,k} = s_{\ell+1,m^{(\ell+1)}(k)} - s_{\ell,k}$, *where* $m^{(\ell+1)}(k)$ *is the index of the level* $\ell+1$ *parent of set* $\mathcal{I}_k^{(\ell)}$.

Note that $\eta_{\ell,k}$ (where $\eta_{\ell,k} \ge 0$ by Assumption 1) is the marginal benefit of fulfillment in level ℓ instead of level $\ell+1$ of any demand occurring in $\mathcal{I}_k^{(\ell)}$. Similarly, $\eta_{L-1} \ge 0$ is the marginal benefit of using a unit of inventory for fulfillment with the highest cost instead of holding onto it. Reformulation of the total cost in this manner is a common technique in newsvendor networks when there is a hierarchy in how demand is fulfilled. For instance, Dong and Rudi (2004) use it for a two-level nested structure. We extend this technique to more general nested structures.

The implication of Lemma 1 is that solving for the worst-case newsvendor cost under a nested fulfillment structure boils down to finding a *tight* bound on the expected cost of unmet demand after each fulfillment level:

$$\mathbb{E}_f\left[\sum_{\ell=0}^{L-1}\eta_\ell^\top(\mathbf{E}_\ell\tilde{\mathbf{D}}-\mathbf{E}_\ell\mathbf{y})^+\right], \quad (7)$$

for any f in the set $\mathcal{F}_{\ge 0}$ defined in (4). We will use this important fact for the rest of the paper.

Even with a closed-form cost function, finding a tight bound on (7) and solving the robust multilocation problem is NP-hard. This is because Hanasusanto et al. (2015) prove that the robust *multi-item* newsvendor problem is NP-hard under mean and covariance information, and any instance of the multi-item newsvendor problem maps to a multilocation newsvendor problem by setting $s_{ij} \to \infty$ for all $i \ne j$ (or sufficiently large enough such that risk pooling is not beneficial).

If the distribution has unrestricted support, the following result shows that a tight bound on (7) is the optimal value of an SDP with an exponential number of semidefinite constraints. (We replace $\sup_{f\in\mathcal{F}_{\ge 0}}$ with $\sup_{f\in\mathcal{F}}$, where we define $\mathcal{F}$ as the set of all distributions with mean $\mathbf{m}$ and covariance $\mathbf{\Sigma}$.)

Lemma 2 (Bertsimas et al. 2010). *For any* $\mathbf{y}\in\Re^n$,

$$\begin{aligned}\sup_{f\in\mathcal{F}}\ \mathbb{E}_f&\left[\sum_{\ell=0}^{L-1}\eta_\ell^\top(\mathbf{E}_\ell\tilde{\mathbf{D}}-\mathbf{E}_\ell\mathbf{y})^+\right]\\ &= \min_{t,\mathbf{r},\mathbf{Y}}\quad t + \mathbf{r}^\top\mathbf{m} + \langle\mathbf{Y},\mathbf{\Sigma}+\mathbf{m}\mathbf{m}^\top\rangle\\ &\quad\text{s.t.}\quad \begin{pmatrix}\mathbf{Y} & \frac{1}{2}(\mathbf{r}-\mathbf{a})\\ \frac{1}{2}(\mathbf{r}-\mathbf{a})^\top & t+\mathbf{a}^\top\mathbf{y}\end{pmatrix}\succeq 0,\quad \forall\mathbf{a}\in\mathcal{L}, \quad (8)\end{aligned}$$

where $\mathcal{L} := \{\mathbf{a}\mid \mathbf{a} = \sum_{\ell=0}^{L-1}\mathbf{E}_\ell^\top(\eta_\ell\odot\mathbf{e}_{A_\ell})$ *for some* $(A_0,A_1,\ldots,A_{L-1})\in 2^{[n_0]}\times 2^{[n_1]}\times\cdots\times 2^{[n_{L-1}]}\}$, $\mathbf{e}_{A_\ell}$ *is an* n_ℓ*-dimensional binary vector where* $e_{A_\ell,k} = 1$ *if and only if* $k\in A_\ell$, *and* $\odot$ *is the element-wise product operator.*

This result can be derived using a vertex-based reformulation (Bertsimas et al. 2010) for minmax stochastic linear programs. We provide a different and simpler proof in the e-companion.

The implication from Lemma 2 is that an exact solution to the robust problem can be found by solving an SDP with 2^N semidefinite constraints, where $N := \sum_{\ell=0}^{L-1} n_\ell$ is the total number of nodes in the tree representation of the nested structure. SDPs can be solved through interior point methods which have polynomial time worst-case complexity (Vandenberghe and Boyd 1996). However, the SDP in (8) is not computationally tractable beyond small values for n, since it involves $\mathcal{O}(2^N)$ constraints that each require a $(n+1)\times(n+1)$ matrix to be positive semidefinite. In the worst case, $N = \frac{n(n+1)}{2}$ if the number of levels is $L = n$ and $n_\ell = n-\ell$.

Due to the complexity, we develop a computationally tractable heuristic for the robust problem under nested costs in Section 5. However, for smaller instances of the problem (two locations), we can obtain the optimal solution analytically in Section 4, which allows us to derive insights about the effects of demand ambiguity on the optimal inventory solution under risk pooling.

4. A Two-Location Model

Even though the general multilocation problem is NP-hard, in this section we study the case of two locations ($n = 2$) with risk pooling. We show that, in this special case, we can derive an exact expression for the robust optimal inventory levels.

In previous works on the classical newsvendor (i.e, $n = 1$), tight closed-form bounds on the expected unmet demand and the expected newsvendor cost have been derived that rely on specific parameters of the demand distribution. The most well-known of these uses mean and variance (Scarf 1958, Gallego and Moon 1993), which we state below for completeness.

Lemma 3 (Scarf 1958, Gallego and Moon 1993). *If* $\tilde{d}$ *has mean* m *and standard deviation* σ, *then for any* $y\in\Re$, $\mathbb{E}(\tilde{d}-y)^+ \le \frac{1}{2}(m-y) + \frac{1}{2}\sqrt{(m-y)^2+\sigma^2}$. *Moreover, this bound is tight since there exists a two-point demand distribution where the bound holds with equality.*

In a single-location model, a bound for $\mathbb{E}(\tilde{d}-y)^+$ yields a bound for the maximal total expected cost, as there are no levels to fulfillment. However, in the case where two locations share inventory ($n = 2$), the Scarf bound cannot reveal how pooling the inventory

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of *multiple correlated locations* impacts the maximal expected cost. Specifically, to obtain a bound to the expected cost for two locations with inventory risk pooling, we need to obtain a bound for

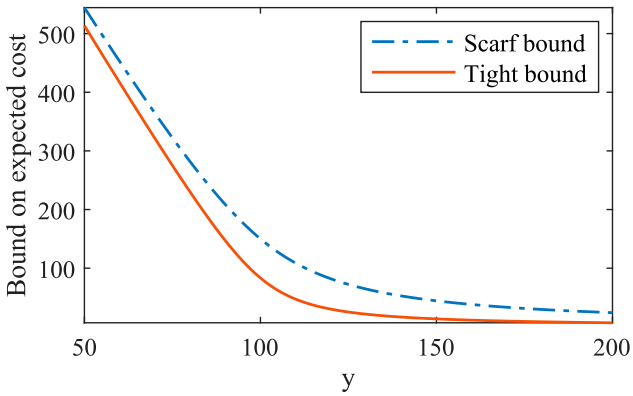
$$\mathbb{E}_f\left[\left(\tilde{d}_1 - y_1\right)^+ + \left(\tilde{d}_2 - y_2\right)^+ + \zeta\left(\tilde{d}_1 + \tilde{d}_2 - y_1 - y_2\right)^+\right], \quad (9)$$

for some parameter $\zeta \geq 0$. Note that the above expression (9) represents the cost of unmet demand after each fulfillment level, where the first two terms correspond to fulfillment at level 0, and the final term corresponds to network fulfillment at level 1.

When the mean and the covariance of $(\tilde{d}_1, \tilde{d}_2)$ are known, the tight bound on (9) can be much smaller than the bound resulting from simply applying the Scarf bound to each term in (9), since the latter approach ignores the relationships of these terms due to demand spillovers. Figure 3 plots the sum of Scarf bounds against the tight bound (found numerically using Lemma 2).

Hence, our aim in this section is to study pooling in a two-location setting ($n = 2$) by deriving a closed-form tight bound on (9) given only mean and covariance information. Let m_i and σ_i be the mean and the standard deviation of demand $\tilde{d}_i$ in location $i = 1, 2$. The demands $\tilde{d}_1, \tilde{d}_2$ are correlated with coefficient $\rho \in [-1, 1]$. As in the previous section, we refer to the set of all joint distributions with these statistics as $\mathcal{F}$. The tight bound on (9) for any $f \in \mathcal{F}$ is presented later in Proposition 1. We first present the idea of its derivation (where the details of the proof are found in Section EC.2.3 of the e-companion).

Figure 3. (Color online) Scarf Bound and Tight Bound on Expected Cost (9) for $n = 2$ with $\zeta = 4, \rho = -1$ and $m_i = 100, \sigma_i = 50$ for $i = 1, 2$



The supremum of (9) in $\mathcal{F}$ is a moment problem which we show to have the dual problem:

$$\begin{aligned} \sup_{t, u_1, u_2, r_1, r_2, v} \quad & t + m_1 r_1 + m_2 r_2 + \left(m_1^2 + \sigma_1^2\right) u_1 \\ & + \left(m_2^2 + \sigma_2^2\right) u_2 + \left(m_1 m_2 + \rho \sigma_1 \sigma_2\right) v \\ \text{s.t.} \quad & g(d_1, d_2;\ t, u_1, u_2, r_1, r_2, v) \leq q(d_1, d_2), \\ & \qquad \forall (d_1, d_2) \in \Re^2. \end{aligned} \quad (10)$$

Here, $q(d_1, d_2)$ is a piecewise linear function in (d_1, d_2) with six pieces (shown in Figure 4(a)), and $g(d_1, d_2; \theta) := t + r_1 d_1 + r_2 d_2 + u_1 d_1^2 + u_2 d_2^2 + v d_1 d_2$ is a parametric quadratic curve in (d_1, d_2) (shown in Figure 4(b)), where the dual variables $\theta := (t, u_1, u_2, r_1, r_2, v)$ are the parameters. A dual solution $\theta = (t, u_1, u_2, r_1, r_2, v)$ is feasible if the quadratic function is bounded above by the piecewise-planar function in all of $\Re^2$. (Figure 4(c) shows these functions corresponding to a dual feasible solution.)

We then find set Θ of dual feasible solutions where any $\theta \in \Theta$ corresponds to a quadratic function $g(d_1, d_2; \theta)$ that touches $q(d_1, d_2)$ in exactly six points (see Figure 4(d)). Note that Θ is not a singleton; hence, we find necessary and sufficient conditions on $\theta = (t, u_1, u_2, r_1, r_2, v)$ that identify if $\theta \in \Theta$. (These conditions are (EC.9)–(EC.13) in the e-companion.) From these conditions, we show that Θ has a parametric form with only a single parameter. We can then find the maximum dual objective of (10) over the dual solutions in Θ by solving an optimization problem on a single parameter. By weak duality, the optimal solution θ^* has a dual value that upper bounds (9) for any $f \in \mathcal{F}$.

To show that this is a *tight* upper bound, we next find a primal feasible solution $f^* \in \mathcal{F}$ whose expectation in (9) matches the bound. We do this by setting the support points of f^* as the intersection points of $g(\cdot; \theta^*)$ with $q(\cdot)$ (shown as the bright points in each face of the piecewise-planar function in Figure 4(d)). Hence, f^* has six support points, and we find the probabilities $\pi_1, \ldots, \pi_6$ that sum up to 1 and where the first two central moments are equal to $(m_1, m_2, \sigma_1^2, \sigma_2^2, \rho\sigma_1\sigma_2)$; on such a distribution, we show that the bound is tight. In fact, there could be many such probability distributions, and (EC.16)–(EC.20) in the e-companion provide necessary and sufficient conditions for the desired properties. To complete the derivation, we need to show that there exists a distribution where the probabilities are nonnegative. Condition 1 below is the necessary and sufficient condition to ensure this.

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Condition 1. *The values of inventory* $\mathbf{y} = (y_1, y_2) \in \Re^2$, *moments* $(m_1, m_2, \sigma_1, \sigma_2, \rho)$, *and parameter* ζ *are such that the interval* $[\max(0, \alpha_1, \alpha_5), \min(\alpha_2, \alpha_3, \alpha_4)]$ *is nonempty, where we define*

$$\alpha_1 := \frac{1}{2(\nu^2-1)}\left[\frac{m_2 - y_2 + \nu(m_1 - y_1)}{\Phi(\mathbf{y};\nu)} - \frac{(m_2-y_2)^2 + \sigma_2^2 + \nu(m_1-y_1)(m_2-y_2) + \nu\rho\sigma_1\sigma_2}{\Phi(\mathbf{y};\nu)^2}\right],$$

$$\alpha_2 := \frac{1}{(\nu^2-1)}\left[\frac{(m_1-y_1)^2+\sigma_1^2}{\Phi(\mathbf{y};\nu)^2} - 1\right],$$

$$\alpha_3 := \frac{1}{2(\nu^2-1)}\left[\frac{(m_1-y_1)^2+(m_2-y_2)^2+\sigma_1^2+\sigma_2^2}{\Phi(\mathbf{y};\nu)^2} - 2\right] + \frac{1}{2(\nu+1)}\left[\frac{m_1-y_1}{\Phi(\mathbf{y};\nu)} - \frac{m_2-y_2}{\Phi(\mathbf{y};\nu)}\right],$$

$$\alpha_4 := \frac{1}{2(\nu^2-1)}\left[\frac{2-\nu}{(\nu-1)}\left(1 - \frac{(m_1-y_1)^2+\sigma_1^2}{\Phi(\mathbf{y};\nu)^2}\right) + \frac{1}{(\nu-1)}\left(\nu^2 - \frac{(m_2-y_2)^2+\sigma_2^2}{\Phi(\mathbf{y};\nu)^2}\right) + \frac{(m_2-y_2+\nu(m_1-y_1))}{\Phi(\mathbf{y};\nu)}\right],$$

$$\alpha_5 := \frac{1}{2(\nu^2-1)}\left[\frac{(m_1-y_1)^2+\sigma_1^2}{\Phi(\mathbf{y};\nu)^2} - \frac{(m_2-y_2)^2+\sigma_2^2}{\Phi(\mathbf{y};\nu)^2}\right] + \frac{1}{2(\nu+1)}\left[\frac{m_1-y_1}{\Phi(\mathbf{y};\nu)} - \frac{m_2-y_2}{\Phi(\mathbf{y};\nu)}\right],$$

for $\nu = (3\zeta+1)/(\zeta+1)$ *and* $\Phi(\mathbf{y};\nu)$ *defined as*

$$\Phi(\mathbf{y};\nu) := \sqrt{\frac{(m_1-y_1)^2+\sigma_1^2 + (m_2-y_2)^2+\sigma_2^2 + (\nu-1)\cdot(\rho\sigma_1\sigma_2 + (m_1-y_1)(m_2-y_2))}{\nu+1}}. \tag{11}$$

When this condition holds, we know that the bound from our derivation is tight since there exists a primal feasible solution $f^* \in \mathcal{F}$ that matches the bound. This is stated formally in the next result.

Proposition 1. *If* $n = 2$, *then for any* $\zeta > 0$, $\mathbf{y} = (y_1, y_2) \in \Re^2$, *and* $f \in \mathcal{F}$,

$$\mathbb{E}_f\left[\left(\tilde{d}_1 - y_1\right)^+ + \left(\tilde{d}_2 - y_2\right)^+ + \zeta\left(\tilde{d}_1 + \tilde{d}_2 - y_1 - y_2\right)^+\right] \le \frac{1}{2}(\zeta+1)(m_1 - y_1 + m_2 - y_2) + (\zeta+1)\Phi\left(\mathbf{y}; \frac{3\zeta+1}{\zeta+1}\right), \tag{12}$$

where $\Phi(\mathbf{y};\nu)$ *is defined in* (11). *Moreover, if Condition* 1 *is satisfied, then this bound is tight for some probability distribution* $f_{\mathbf{y}}^* \in \mathcal{F}$ *with at most six support points.*

We can now use Lemma 1 to get the worst-case expected newsvendor cost when $n = 2$. Consider a two-location newsvendor network where the in-location fulfillment cost is equal to s_0. Furthermore, it is possible to do cross-location fulfillment at a higher cost s. As before, p is the penalty cost for lost sales, and h is the cost of excess inventory. The following theorem presents the tight upper bound on the expected newsvendor cost if $f \in \mathcal{F}$.

Theorem 1. *If* $n = 2$ *and* $p + h > s > s_0$, *then for any* $\mathbf{y} = (y_1, y_2) \in \Re^2$,

$$\sup_{f\in\mathcal{F}} \mathbb{E}_f[C(\mathbf{y}, \tilde{\mathbf{D}})] \le s_0(m_1+m_2) - \frac{1}{2}(p-h-s_0)(y_1+y_2-m_1-m_2) + (p+h-s_0)\Phi\left(\mathbf{y}; \frac{3(p+h-s)+s-s_0}{p+h-s_0}\right), \tag{13}$$

where $\Phi(\cdot;\nu)$ *is defined in* (11). *Moreover, if Condition* 1 *is satisfied with* $\zeta = \frac{p+h-s}{s-s_0}$, *then this bound is tight for some probability distribution* $f_{\mathbf{y}}^* \in \mathcal{F}$ *with at most six support points.*

Note the importance of Condition 1 in ensuring that the bound (13) on the expected newsvendor cost is tight. Although this condition appears complicated, it is straightforward to check whether any given value for (y_1, y_2) satisfies Condition 1. However, the following lemma provides a class of solutions (y_1, y_2) where Condition 1 holds.

Lemma 4. *Condition* 1 *is satisfied if* $y_1 - m_1 = y_2 - m_2$ *and if*

$$\frac{\sigma_1^2+\sigma_2^2}{(\nu^2+1)} \le \frac{\sigma_1^2+\sigma_2^2+(\nu-1)\rho\sigma_1\sigma_2}{\nu+1} \le \min(\sigma_1^2, \sigma_2^2), \tag{14}$$

where $\nu = (3\zeta+1)/(\zeta+1)$.

Lemma 4 states that, under (14), Condition 1 holds when $y_1 - m_1 = y_2 - m_2$. In fact, Condition 1 may still hold even when $y_1 - m_1 \neq y_2 - m_2$. But Lemma 4 is important since it allows us to find the inventory solution (y_1^*, y_2^*) that minimizes the maximal expected newsvendor cost. This is because the minimizer $(\bar{y}_1, \bar{y}_2)$ of the upper bound function (13) has the property that $\bar{y}_1 - m_1 = \bar{y}_2 - m_2$, which satisfies Condition 1 (by Lemma 4). Hence, Theorem 1 and the convexity of the worst-case expected newsvendor cost imply that $(y_1^*, y_2^*) = (\bar{y}_1, \bar{y}_2)$. This result is stated formally in the following proposition, whose proof is in the e-companion.

Proposition 2. *If* $n = 2$ *and if* $(\sigma_1, \sigma_2, \rho)$ *satisfy condition* (14) *with* $\nu = \frac{3(p+h-s)+s-s_0}{p+h-s_0}$, *then the inventory level that*

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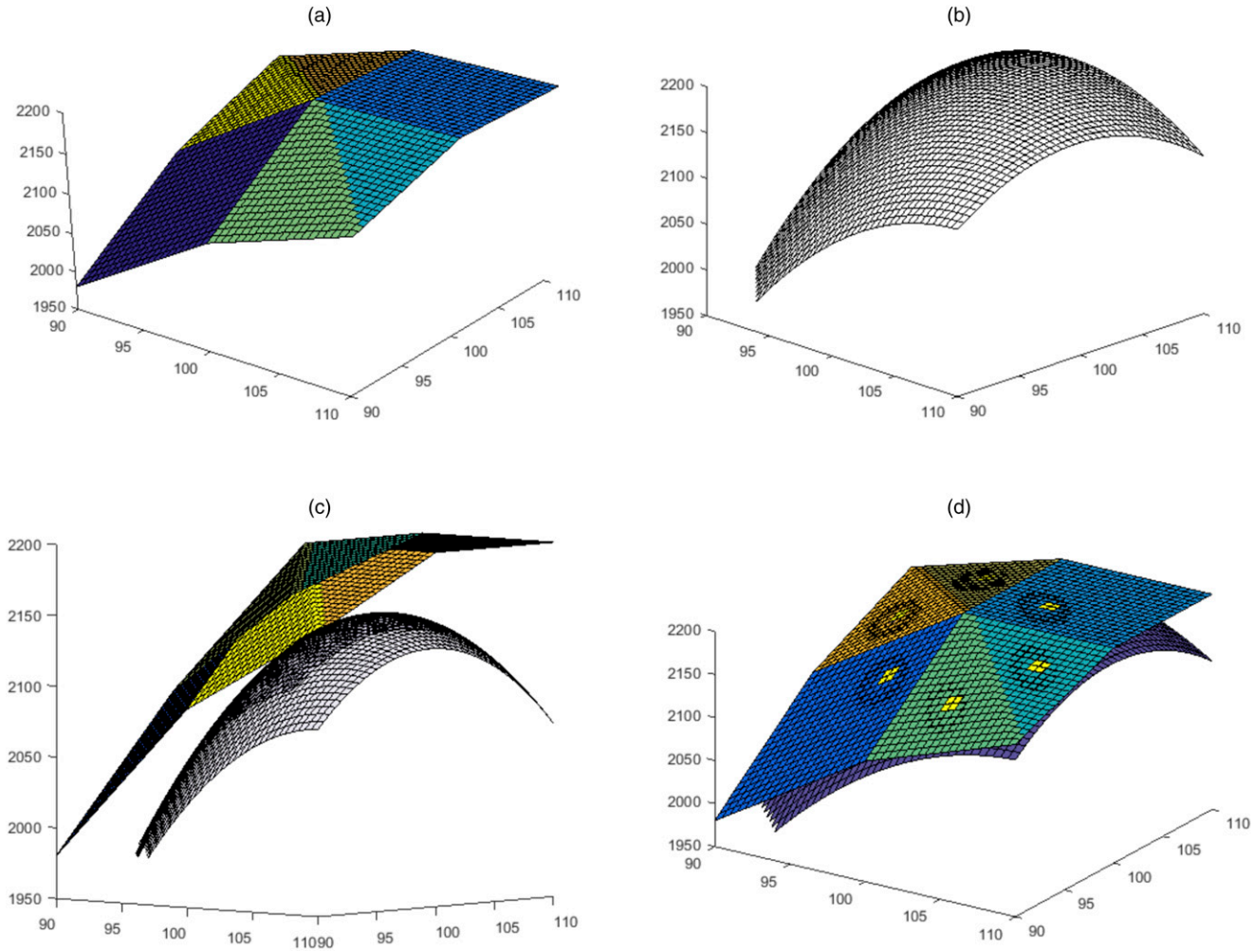
Figure 4. (Color online) Illustration of the Dual Program: (a) The Piecewise-Planar Function $q(d_1, d_2)$; (b) The Quadratic Function $g(d_1, d_2)$; (c) The Functions Corresponding to a Dual Feasible Solution; (d) The Functions Corresponding to the Dual Optimal Solution



minimizes the maximal expected cost over distributions in $\mathcal{F}$ *is* $\mathbf{y}^* = (y_1^*, y_2^*)$, *where*

$$y_1^* - m_1 = y_2^* - m_2 = \frac{1}{2}(p - h - s_0) \times \sqrt{\frac{(p + h - s_0)(\sigma_1^2 + \sigma_2^2) + 2(p + h - s)\rho\sigma_1\sigma_2}{2h(2(p + h) - s - s_0)(p - s_0)}}.$$

The minmax expected cost is

$$C^* = s_0(m_1 + m_2) + \sqrt{\frac{2h(p - s_0)\left[(p + h - s_0)(\sigma_1^2 + \sigma_2^2) \quad + 2(p + h - s)\rho\sigma_1\sigma_2\right]}{2(p + h) - s - s_0}},$$

which is achieved by a six-point support joint distribution.

Note that if cross-location fulfillment is not allowed, then each location will set a safety stock equal to what is prescribed by Scarf (1958), so $\tilde{y}_i - m_i = (p - h - s_0)\sigma_i/(2\sqrt{h(p - s_0)})$ for $i = 1, 2$. It is easy to check from Proposition 2 that $|y_1^* - m_1| + |y_2^* - m_2| \le |\tilde{y}_1 - m_1| + |\tilde{y}_2 - m_2|$. Hence, an insight from Proposition 2 is that inventory risk sharing results in a smaller safety stock magnitude, mirroring a similar insight from stochastic systems.

4.1. Discussion of the Worst-Case Distribution

Theorem 1 states that, for any inventory level $\mathbf{y} = (y_1, y_2)$ satisfying Condition 1, the worst-case demand distribution has at most six support points. We provide in Table EC.3 of the e-companion the expressions for the support points and the probabilities of this worst-case distribution $f_{\mathbf{y}}^*$. (The subscript $\mathbf{y}$ is to emphasize that the worst-case distribution depends on the inventory level.) Figure 5 illustrates these support points, where $\nu = \frac{3(p+h-s)+s-s_0}{p+h-s_0}$ and $\Phi(\mathbf{y}; \nu)$ is defined in (11).

The figure shows each support point lying on one of three solid lines that pass through (y_1, y_2). Note that the mean (m_1, m_2) (not shown in the figure) is in the

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convex hull of the six points where (y_1, y_2) is the center. The six points stretch away (in the direction indicated by the arrows) as $\Phi(\mathbf{y}; \nu)$ grows, which occurs when we increase $|y_i - m_i|$, increase σ_i, or increase ρ. The dashed line $(d_1 + d_2 = y_1 + y_2)$ corresponds to a perfect balance between demand and inventory after demand spillover. Two of the solid lines converge to this dashed line as $\nu \to 1$, which occurs when $h + p$ and s approach each other. Thus, when the overage or the underage cost is high (or the fulfillment cost s is low), the solid lines pivot further away from the dashed line, and hence, the support points of the worst-case distribution result in very large excess inventory or unmet demand after pooling.

We call a system imbalanced if, after in-location fulfillment, there is leftover inventory in one location and unfulfilled demand at the other location. In Figure 5, any demand realization in the shaded regions induces a system imbalance. Note that realizations in these regions that do not lie on the dashed line lead to high costs—not only are items shipped across locations at higher costs than in-location fulfillment, they also lead to excess leftover inventory or unmet demand at one of the locations. We say that location 1 is imbalanced if it has excess inventory even after helping location 2 (when $y_1 > d_1 + d_2 - y_2 > d_1$) or if it has unmet demand even after help from location 2 (when $d_1 > y_1 + y_2 - d_2 > y_1$). From the worst-case probabilities, location 1 is imbalanced in two of the six outcomes, with total probability

$$\frac{1}{\nu^2 - 1} \cdot \left[\frac{(m_1 - y_1)^2 + \sigma_1^2}{\Phi(\mathbf{y})^2} - 1 \right]. \tag{15}$$

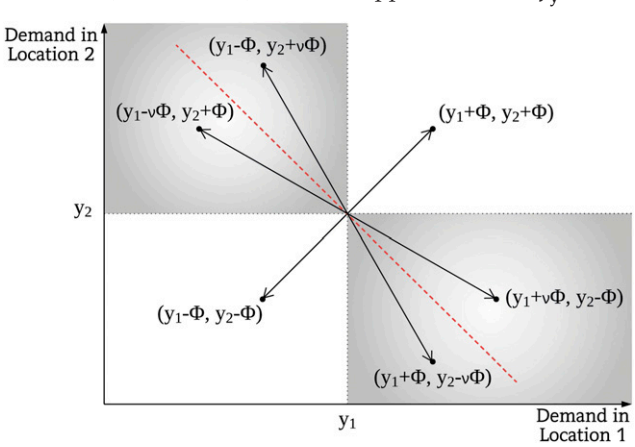
Likewise, location 2 is imbalanced if it has excess inventory even after helping location 1 (when $y_2 > d_1 + d_2 - y_1 > d_2$) or if it has unmet demand even after help from location 1 (when $d_2 > y_1 + y_2 - d_1 > y_2$). The total probability of this occurring (from two outcomes) is

$$\frac{1}{\nu^2 - 1} \cdot \left[\frac{(m_2 - y_2)^2 + \sigma_2^2}{\Phi(\mathbf{y})^2} - 1 \right]. \tag{16}$$

Observe that if the newsvendor chooses inventory levels $\mathbf{y} = (y_1, y_2)$ with nonidentical safety stocks (i.e., either $y_1 - m_1 > y_2 - m_2$ or $y_2 - m_2 > y_1 - m_1$), then (15) and (16) show that the "adversary" will react by putting a higher *probability* on outcomes where the location with higher safety stock is imbalanced. Moreover, the *costs* associated with these imbalances are large when safety stocks are nonidentical; this is because the magnitudes of imbalances increase with $\Phi(\mathbf{y})$, which scales the distance of all support points from (y_1, y_2). However, for any fixed *total* safety stock K (i.e., $y_1 - m_1 + y_2 - m_2 = K$), $\Phi(\mathbf{y})$ is smallest when the safety stock is equally distributed between the locations (i.e., $y_1 - m_1 = y_2 - m_2 = K/2$). Hence, these observations reveal that if the newsvendor decides to place a larger safety stock in one location, the adversary will react by increasing both the *probability* that this location is imbalanced, as well as the *costs* associated with imbalances (i.e., negating the effectiveness of the larger safety stock). Hence, the best strategy against such an adversary is to set the same safety stock in the locations so that $y_1^* - m_1 = y_2^* - m_2$ (Proposition 2).

Interestingly, the equal safety stocks in the optimal robust solution are in contrast to when the demands have a joint normal distribution and the optimal safety stock is proportional to the standard deviation (Dong and Rudi 2004, lemma 1). Hence, our analysis reveals that the strategy of the newsvendor must be different when the distribution is ambiguous.

Figure 5. (Color online) The Six Support Points of $f_{\mathbf{y}}^*$



Notes. Each support point lies on one of three solid lines passing through (y_1, y_2), where the distance from this point increases in proportion to $\Phi(\mathbf{y}; \nu)$. The dashed line corresponds to a perfect balance between demand and supply after demand spillover.

4.2. Insights on the Benefit of Pooling

We now discuss how a distributionally robust framework can be used to analyze the benefit of pooling. In particular, note that if the locations have inventory levels (y_1, y_2), then the benefit of pooling under a joint distribution f is exactly

$$B_f(y_1, y_2) := (p + h - s) \cdot \mathbb{E}_f\left[\left(\tilde{d}_1 - y_1\right)^+ + \left(\tilde{d}_2 - y_2\right)^+ - \left(\tilde{d}_1 + \tilde{d}_2 - y_1 - y_2\right)^+ \right].$$

In other words, it is proportional to the expected reduction in the unmet demand after pooling. However, this benefit can only be determined if we know the true demand distribution f. Knowing only the mean vector and covariance matrix, we want to know whether there is a tight upper bound on the benefit of pooling, which we can get by solving $\sup_{f \in \mathcal{F}} B_f(y_1, y_2)$.

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Note that we cannot use Proposition 1 to find this bound since it requires $\zeta > 0$. In Section EC.3 of the e-companion, we show how we can determine this bound numerically by solving a SDP.

In Figure 6, we plot the tight bound on the benefit of pooling $\sup_{f\in\mathcal{F}} B_f(y)$ as a function of y, where $y_1 = y_2 = y$. (The inventory levels are equal because we choose $m_1 = m_2 = 100$, $\sigma_1 = \sigma_2 = 50$.) The figure shows the results for different values of ρ, where we normalize $p + h - s$ to 1. It is well known that, under specific demand distributions, the benefit of pooling is generally decreasing in the correlation coefficient ρ. However, it is evident from Figure 6 that the maximum benefit from pooling is nonmonotone in ρ, specifically in the region where y is close to the mean demand. Hence, a distributionally robust approach to the risk pooling problem leads to new insights on optimal inventory levels and the benefits of pooling.

5. Heuristic for a Network with Nested Fulfillment Costs

Solving the general multilocation newsvendor problem is NP-hard, and only in the simplest case (two locations) do we have a closed-form tight solution. The challenge is that there are multiple demands which interact, not only because they are correlated, but also because of inventory pooling. Therefore, for general networks, the best that we can hope for is a computationally tractable heuristic to solve the distributionally robust problem. In this section, we develop such a heuristic for nested fulfillment structures by developing a computationally tractable upper bound on (7) for any distribution in $\mathcal{F}_{\geq 0}$. Importantly, this upper bound preserves the cascading nature of the fulfillment induced by the nested costs, resulting in a tighter bound than by simply using the sum of Scarf bounds (26). Note that even if the fulfillment costs do not have a nested structure, we can use our hierarchical agglomerative clustering technique (Section 2.1) to get a nested cost approximation which, combined with the results of this section, yields a computationally tractable heuristic for more general cost structures.

The upper bound result is presented later in Proposition 3. We first discuss the sequence of reformulations (depicted in Figure 7) to derive the upper bound. Introducing $\mathbf{x}^{(\ell)}$ as a n_ℓ-dimensional binary vector, for $\ell = 0, 1, \ldots, L-1$, we can reformulate (7) as follows:

$$\mathbb{E}_f\left[\max_{\substack{\mathbf{x}^{(0)}\in\{0,1\}^{n_0},\\ \mathbf{x}^{(1)}\in\{0,1\}^{n_1},\ldots,\\ \mathbf{x}^{(L-1)}\in\{0,1\}^{n_{L-1}}}} \sum_{\ell=0}^{L-1}\sum_{k=1}^{n_\ell} \eta_{\ell,k}\cdot x_k^{(\ell)}\cdot\left(\mathbf{E}_\ell\tilde{\mathbf{D}} - \mathbf{E}_\ell\mathbf{y}\right)_k\right]. \quad (17)$$

Since $\eta_{\ell,k} \geq 0$, for a given demand realization $\mathbf{D}$, the maximization problem inside (17) will choose $x_k^{(\ell)} = 1$ if and only if the kth element of vector $\mathbf{E}_\ell\mathbf{D} - \mathbf{E}_\ell\mathbf{y}$ is greater than or equal to zero. Hence, the optimal value of the inner problem in (17) is equal to $\sum_{\ell=0}^{L-1}\eta_\ell^\top(\mathbf{E}_\ell\mathbf{D} - \mathbf{E}_\ell\mathbf{y})^+$, so (7) and (17) are equivalent.

Our restriction to a nested fulfillment cost is crucial since it relates the cost of a multilocation newsvendor to the sum of piecewise linear functions (7) (see Lemma 1). Hence, the newsvendor cost is connected

Figure 6. (Color online) An Upper Bound on the Benefit of Pooling with Two Locations Having $y_1 = y_2 = y$ for Different Values of ρ, Where $m_1 = m_2 = 100$, $\sigma_1 = \sigma_2 = 50$, and $p + h - s = 1$

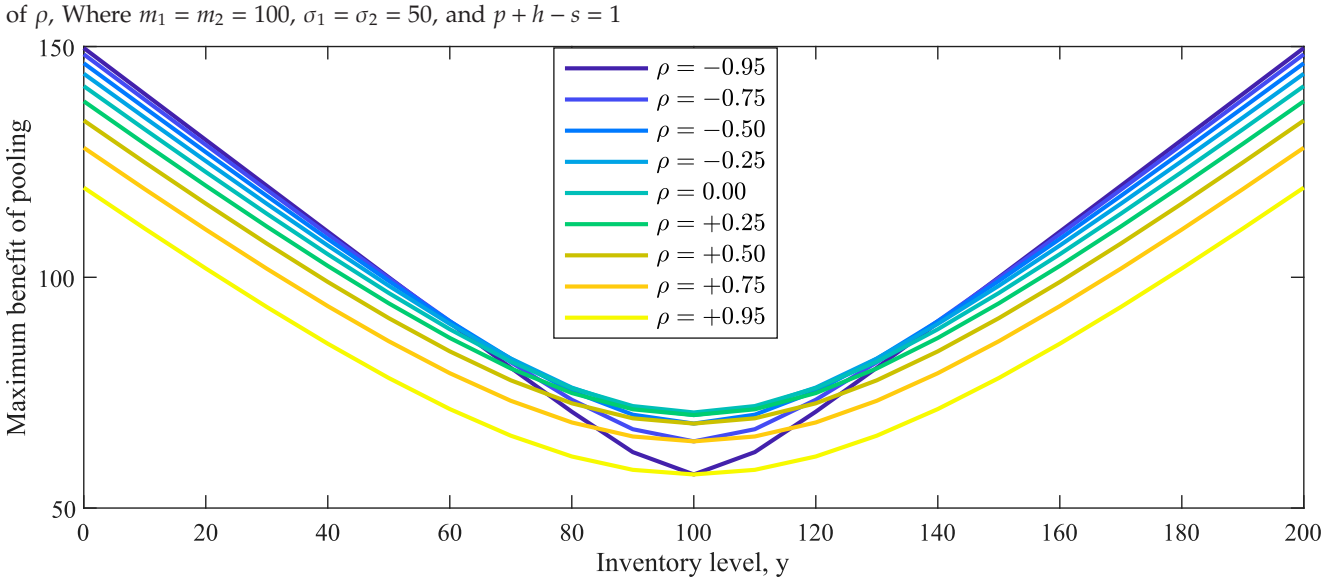

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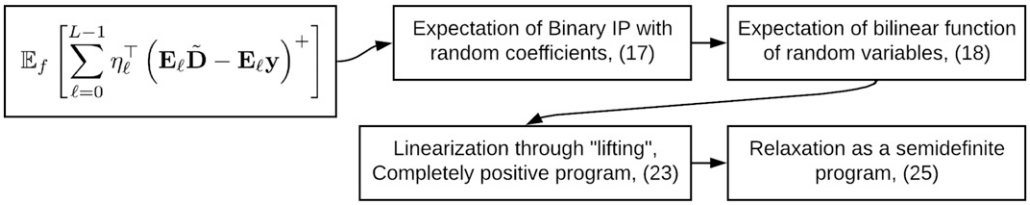
Figure 7. Flowchart of Reformulations



to a 0-1 linear program with a random objective coefficient vector (17). This connection allows us to utilize a reformulation technique in (Natarajan et al. 2011, theorem 3.3). Specifically, since the value of the maximizer in (17) depends on the realization of random demands, the maximizers are random variables, which we denote as $\{\tilde{\mathbf{x}}^{(\ell)}\}_{\ell=0}^{L-1}$. Hence, (17) is equivalent to

$$\mathbb{E}_f\left[\sum_{\ell=0}^{L-1}\sum_{k=1}^{n_\ell}\eta_{\ell,k}\cdot\tilde{x}_k^{(\ell)}\cdot\left(\mathbf{E}_\ell\tilde{\mathbf{D}}-\mathbf{E}_\ell\mathbf{y}\right)_k\right], \tag{18}$$

where we use the tilde on the binary variables to emphasize that they are stochastic variables. In fact, reformulations (17)–(18) preserve the hierarchical structure of the network, as we discuss later in Section 5.1. Specifically, there is an inherent interdependence among the values in $\{\tilde{\mathbf{x}}^{(\ell)}\}_{\ell=0}^{L-1}$ due to the nested structure of the fulfillment network.

Note that reformulation (18) has cross products of random variables, where the underlying uncertainty has a joint distribution f with nonnegative support and with mean $\mathbf{m}$ and covariance $\mathbf{\Sigma}$. A technique to relax a bilinear function into a linear formulation is to introduce new variables, which *lifts* the problem to a higher dimensional space.

Consider the N-dimensional random vector $\tilde{\mathbf{x}} := \left(\tilde{\mathbf{x}}^{(L-1)\top}\ \ \tilde{\mathbf{x}}^{(L-2)\top}\ \ \cdots\ \ \tilde{\mathbf{x}}^{(0)\top}\right)^\top$. We linearize the function (18) by lifting it to a higher dimensional space by introducing the following new variables:

$$\mathbf{x} := \mathbb{E}_f(\tilde{\mathbf{x}}) \in \mathfrak{R}^N, \tag{19}$$

$$\mathbf{Q} := \mathbb{E}_f\left(\tilde{\mathbf{x}}\tilde{\mathbf{D}}^\top\right) \in \mathfrak{R}^{N\times n}, \tag{20}$$

$$\mathbf{R} := \mathbb{E}_f\left(\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top\right) \in \mathfrak{R}^{N\times N}. \tag{21}$$

Defining $N_{L-1} := 0$ and $N_\ell := \sum_{m=0}^{L-\ell-2} n_{L-1-m}$ for $\ell = 0, 1, \ldots, L-2$, we now have that (18) is equivalent to the linear function

$$\sum_{\ell=0}^{L-1}\sum_{k\in[n_\ell]}\eta_{\ell,k}\cdot\left(\sum_{i\in\mathcal{G}_k^{(\ell)}}Q_{N_\ell+k,i}-\sum_{i\in\mathcal{G}_k^{(\ell)}}x_{N_\ell+k}\cdot y_i\right). \tag{22}$$

Therefore, an upper bound to (7) can be found by taking the maximum of the linear function (22) with respect to the variables $(\mathbf{x}, \mathbf{Q}, \mathbf{R})$ that satisfy some feasibility constraints which are consistent with (19)–(21). Specifically, since $\tilde{\mathbf{D}}$ and $\tilde{\mathbf{x}}$ are nonnegative, we have that

$$\mathbb{E}_f\left(\begin{pmatrix}1\\ \tilde{\mathbf{D}}\\ \tilde{\mathbf{x}}\end{pmatrix}\begin{pmatrix}1\\ \tilde{\mathbf{D}}\\ \tilde{\mathbf{x}}\end{pmatrix}^\top\right) = \begin{pmatrix}1 & \mathbf{m}^\top & \mathbf{x}^\top\\ \mathbf{m} & \sum + \mathbf{m}\mathbf{m}^\top & \mathbf{Q}^\top\\ \mathbf{x} & \mathbf{Q} & \mathbf{R}\end{pmatrix}$$

is a completely positive matrix. A matrix $\mathbf{A}$ is completely positive (we write $\mathbf{A} \succeq_{cp} 0$) if and only if there exist nonnegative vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ such that $\mathbf{A} = \sum_{i=1}^k \mathbf{v}_i\mathbf{v}_i^\top$. Furthermore, since $\tilde{\mathbf{x}}$ is a vector of binary random variables, we have $R_{ii} = \mathbb{E}[\tilde{\mathbf{x}}_i^2] = \mathbb{E}[\tilde{\mathbf{x}}_i] = x_i$. Hence, we get

$$\begin{aligned}
&\sup_{f\in\mathcal{F}_{\geq 0}}\mathbb{E}_f\left[\sum_{\ell=0}^{L-1}\eta_\ell^\top\left(\mathbf{E}_\ell\tilde{\mathbf{D}}-\mathbf{E}_\ell\mathbf{y}\right)^+\right]\\
&\quad= \max_{\mathbf{x},\mathbf{Q},\mathbf{R}}\ \sum_{\ell=0}^{L-1}\sum_{k\in[n_\ell]}\eta_{\ell,k}\cdot\left(\sum_{i\in\mathcal{G}_k^{(\ell)}}Q_{N_\ell+k,i}-\sum_{i\in\mathcal{G}_k^{(\ell)}}x_{N_\ell+k}\cdot y_i\right)\\
&\qquad\text{s.t.}\ \begin{pmatrix}1 & \mathbf{m}^\top & \mathbf{x}^\top\\ \mathbf{m} & \sum + \mathbf{m}\mathbf{m}^\top & \mathbf{Q}^\top\\ \mathbf{x} & \mathbf{Q} & \mathbf{R}\end{pmatrix}\succeq_{cp} 0,\\
&\qquad\qquad R_{ii} = x_i, \qquad i\in[N].
\end{aligned} \tag{23}$$

This optimization model is known as a *completely positive program* (Abraham and Naomi 2003). Although our arguments above only prove that the left-hand side of (23) is upper bounded by the right-hand side, Natarajan et al. (2011) prove that the bound is tight.

The problem of checking if a matrix is copositive is NP-hard (Dickinson and Gijben 2014). However, there is a stream of literature on approximating completely positive programs with SDPs through a hierarchy of linear and semidefiniteness constraints. Natarajan et al. (2011) use the simple relaxation of replacing $\mathbf{A} \succeq_{cp} 0$ with $\mathbf{A} \succeq 0$ and $\mathbf{A} \geq 0$ to conduct computational experiments. In a follow-up paper, Natarajan and Teo (2017) develop a new SDP formulation by showing that the complexity of the problem is related to the complexity of characterizing the convex hull of rank-one matrices. Natarajan and Teo (2017) and Natarajan et al. (2017) use this approach to develop SDP-based heuristics for the multi-item newsvendor problem (i.e., each inventory node is dedicated to fulfill only one demand node). We next show that this approach can also be applied to the multilocation newsvendor problem if the fulfillment cost structure is nested.

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Following from the fact that $\tilde{\mathbf{x}} \in \{0,1\}^N$, note that we have

$$\begin{pmatrix} 1 & \mathbf{x}^\top \\ \mathbf{x} & \mathbf{R} \end{pmatrix} := \mathbb{E}_f\left(\begin{pmatrix} 1 \\ \tilde{\mathbf{x}} \end{pmatrix}\begin{pmatrix} 1 \\ \tilde{\mathbf{x}} \end{pmatrix}^\top\right) \in \operatorname{conv}\left\{\begin{pmatrix} 1 \\ \mathbf{w} \end{pmatrix}\begin{pmatrix} 1 \\ \mathbf{w} \end{pmatrix}^\top : \mathbf{w} \in \{0,1\}^N\right\}. \tag{24}$$

That is, the left-hand side matrix is a convex combination of a set of Boolean matrices where each matrix is itself a product of Boolean variables. The convex hull of such matrices is referred to as the Boolean quadric polytope (Padberg 1989). Computing the convex hull is a difficult problem since unconstrained binary quadratic programming is NP-hard in general. We instead use a simple linear relaxation of this polytope. Note that if the matrix on the left-hand side of (24) belongs to the Boolean quadric polytope, then $\mathbf{R} \geq \mathbf{0}$ and, for all $i, j \in [N]$, $R_{ii} = x_i$, $R_{ij} \leq x_i$, and $R_{ij} \geq x_i + x_j - 1$.

Proposition 3. *If $f \in \mathcal{F}_{\geq 0}$ where $\mathbf{\Sigma} \succ 0$, then for any $\mathbf{y} \in \mathfrak{R}^n$,*

$$\begin{aligned}
&\mathbb{E}_f\left[\sum_{\ell=0}^{L-1} \eta_\ell^\top (\mathbf{E}_\ell \tilde{\mathbf{D}} - \mathbf{E}_\ell \mathbf{y})^+\right] \\
&\leq \max_{\mathbf{x},\, \mathbf{Q}\geq 0,\, \mathbf{R}\geq 0} \sum_{\ell=0}^{L-1} \sum_{k\in[n_\ell]} \eta_{\ell,k}\left(\sum_{i\in\mathcal{G}_k^{(\ell)}} Q_{N_\ell+k,i} - \sum_{i\in\mathcal{G}_k^{(\ell)}} x_{N_\ell+k}\cdot y_i\right), \\
&\quad \text{s.t.} \quad \begin{pmatrix} 1 & \mathbf{m}^\top & \mathbf{x}^\top \\ \mathbf{m} & \sum + \mathbf{m}\mathbf{m}^\top & \mathbf{Q}^\top \\ \mathbf{x} & \mathbf{Q} & \mathbf{R} \end{pmatrix} \succeq 0, \\
&\qquad R_{ii} = x_i, \qquad i \in [N], \\
&\qquad R_{ij} \leq x_i, \qquad i \in [N], j \in [N], \\
&\qquad R_{ij} \geq x_i + x_j - 1, \qquad i \in [N], j \in [N].
\end{aligned} \tag{25}$$

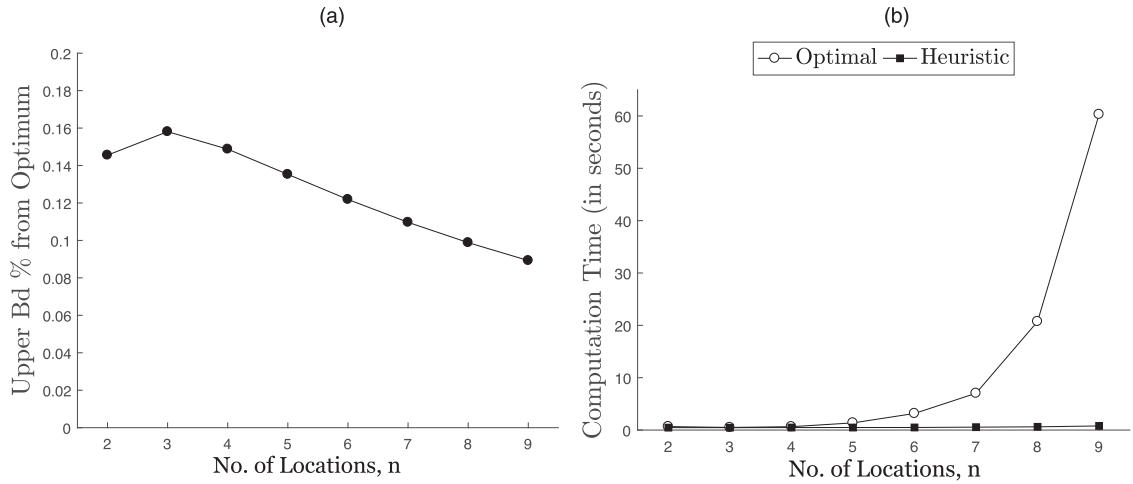
We observe that the only difference between the SDP formulation in (25) and the simple SDP relaxation of (23) is that the former has added constraint types: $R_{ij} \leq x_i$ and $R_{ij} \geq x_i + x_j - 1$. In computational experiments, we find that the gap between the two formulations is generally small (varying from almost zero to 2.5%). The gap is largest whenever the network is small and when inventory levels are low (so demand is more likely to spillover to other locations).

Proposition 3 provides a computationally tractable upper bound on the expected cost of unmet demand under a nested structure. It is tractable since it solves an SDP whose size is polynomial in n. Thus, the implication from Lemma 1 and Proposition 3 is that the robust multilocation newsvendor problem with nested costs can be solved heuristically through a computationally tractable SDP. Section EC.1.3 of the e-companion describes in detail the algorithm that returns the heuristic solution (denoted as $\hat{\mathbf{y}}$) and the approximate minmax robust cost (denoted as $\hat{C}$).

Whether the heuristic returns values $(\hat{\mathbf{y}}, \hat{C})$ that are close approximations to the true robust solution and optimal cost $(\mathbf{y}^*, C^*)$ depends on whether the SDP (25) can tightly approximate (7). We test this computationally by computing the exact tight bound on (7), which by Lemma 2 we can solve exactly when n is small. Since there is no support restriction in Lemma 2, we allow negative values for $\mathbf{q}$ and $\mathbf{Q}$ in (25). Our experiment is on a simple two-level structure. Figure 8 shows the results when the number of locations is varied, with $h = 1$, $p = 100$, $s = 1$, and $s_0 = 0$, with identical marginal distributions (mean $m = 100$ and standard deviation $\sigma = 50$), and with each pair of locations having a correlation coefficient of $\rho = 0.25$.

Figure 8(a) shows the gap between the minmax cost C^* and $\hat{C}$. The plot shows that $\hat{C}$ is close to the minmax cost (within 0.2%), and in general, this gap decreases

Figure 8. Experiments to Evaluate the SDP Heuristic



Notes. (a) The relative gap of C^* and $\hat{C}$. (b) The computation time in seconds of the optimal bound computed from Lemma 2 and the heuristic.

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with the number of locations. Therefore, the upper bound (25) is empirically tight in the neighborhood of $\hat{\mathbf{y}}$ and $\mathbf{y}^*$. This reveals that our proposed solution method provides a good heuristic for approximating the robust inventory levels in a nested cost structure. Figure 8(b) shows the efficiency of our heuristic compared with solving for the robust inventory levels through Lemma 2. We observe that the computational time of the heuristic is significantly smaller compared with the exponentially sized optimal SDP. Even for $n = 9$, the latter required around 20,000 unique SDP variables involved in the SDP constraints, whereas the heuristic required only 20 unique SDP variables, and the disparity is clearly seen in the computational times. The heuristic could solve up to $n = 100$ in under an hour, whereas the optimal solution could not be evaluated even for $n = 10$ due to memory constraints. As a result, our proposed heuristic solves a general class of problems with nested fulfillment cost structures, with significantly less computational burden than the optimal SDP.

5.1. Comparing with Other Computationally Tractable Bounds

Observe that the formulation (17) preserves the hierarchical order of fulfillment since the implied optimization problem inside the square brackets finds a solution $\{\mathbf{x}^{(\ell)}(\mathbf{D})\}_{\ell=0}^{L-1}$ of binary variables that is consistent with the hierarchical fulfillment for a particular demand realization $\mathbf{D}$. To see this, note that since $\eta_{\ell,k} \geq 0$ for any ℓ and $k \in n_\ell$, we have that $x_k^{(\ell)}(\mathbf{D}) = 0$ if and only if $\mathbf{e}_{\ell,k}^\top \mathbf{D} < \mathbf{e}_{\ell,k}^\top \mathbf{y}$. Hence, $x_k^{(\ell)}(\mathbf{D}) = 0$ will be chosen only if there is inventory remaining after level ℓ fulfillment of locations in $\mathcal{I}_k^{(\ell)}$. This can only occur if there exists at least one path in the tree from level 0 to level ℓ (for example, the path through $\{\mathcal{I}_{k_0}^{(0)}, \mathcal{I}_{k_1}^{(1)}, \ldots, \mathcal{I}_{k_\ell}^{(\ell)}\}$) where there is excess inventory in all sets in this path. This makes sense because level ℓ fulfillment will not occur unless there was excess inventory after all the less costly lower level fulfillments. Mathematically, if $x_k^{(\ell)}(\mathbf{D}) = 0$, then there must exist at least one path where $x_{k_0}^{(0)}(\mathbf{D}) = x_{k_1}^{(1)}(\mathbf{D}) = \cdots = x_{k_{\ell-1}}^{(\ell-1)}(\mathbf{D}) = 0$. The inherent interdependence among the values in $\{\mathbf{x}^{(\ell)}(\mathbf{D})\}_{\ell=0}^{L-1}$ is due to the nested structure of the fulfillment costs described by $\Xi = \{\mathbf{E}_\ell\}_{\ell=0}^{L-1}$. Therefore, there must also be interdependence among the values of $\mathbf{x} = \mathbb{E}(\mathbf{x}(\tilde{\mathbf{D}}))$ due to the nested costs. Consider a given path on the fulfillment tree from k_0 (in level 0) to k_1 (in level 1), continuing on until k_ℓ in level ℓ. In computational experiments, we observe that, in the optimal solution to (25), we have $x_{k_0}^{(0)} \geq x_{k_1}^{(1)} \geq \cdots \geq x_{k_{\ell-1}}^{(\ell-1)}$. This discussion serves to show that the SDP (25) preserves information on how demand spills over in the network.

In what follows, we compare the SDP bound (25) against the sum of Scarf bounds. Specifically, using the Scarf bound (Lemma 3), we get

$$\eta_{\ell,k} \cdot \mathbb{E}\left(\mathbf{e}_{\ell,k}^\top \tilde{\mathbf{D}} - \mathbf{e}_{\ell,k}^\top \mathbf{y}\right)^+ \leq \frac{\eta_{\ell,k}}{2}\left(\mathbf{e}_{\ell,k}^\top \mathbf{m} - \mathbf{e}_{\ell,k}^\top \mathbf{y} + \sqrt{\left(\mathbf{e}_{\ell,k}^\top \mathbf{m} - \mathbf{e}_{\ell,k}^\top \mathbf{y}\right)^2 + \mathbf{e}_{\ell,k}^\top \Sigma \mathbf{e}_{\ell,k}}\right), \tag{26}$$

where $\mathbf{e}_{\ell,k}^\top$ is the kth row of matrix $\mathbf{E}_\ell$. Summing the right-hand side of (26) for all ℓ, k is in an upper bound for (7). Unlike our approach, the Scarf bound does not directly account for demand spillovers in the network. Hence, we expect it to be a looser bound than the SDP bound.

We compare the bounds in numerical experiments. We do this by solving the distributionally robust multilocation newsvendor problem, but replacing the worst-case unmet demand with either the SDP bound or the Scarf bound; we refer to its optimal value as the optimal bound. We check whether the SDP approach results in a tighter bound by comparing the reduction in its optimal bound relative to the optimal bound under the Scarf approach. For this experiment, the hierarchical cost structure is such that, in level $\ell = 0$, components are each individual location, then in $\ell = 1$ components are pairs of locations, then in $\ell = 2$ components are four locations, and so on. We set m_i to be randomly chosen between 100 and 200. Standard deviation at i is equal to m_i, and correlation is randomly generated. The fulfillment costs in level 0 are randomly generated in $[1, 3]$, in level 1 they are in $[3, 9]$, and in level 2 they are $[9, 27]$. We set $p = 54$ and $h = 1$.

Table 1 shows, for different values of n, summary statistics of the results from 100 randomly generated problem instances. Based on this table, we observe that the Scarf bound can be a poor approximation of the total expected unmet demand especially for larger network sizes.

6. Numerical Analysis

We next conduct experiments on the proposed heuristic solution based on the SDP bound in Proposition 3.

Table 1. Percentage Reduction in the Optimal Bound when Using the SDP Approach Instead of a Sum of Scarf Bounds: Summary Statistics from 100 Randomly Generated Problem Instances

	$n = 2$	$n = 4$	$n = 8$	$n = 16$
25% quantile	0.00%	0.17%	2.36%	5.47%
Median	0.00%	2.02%	4.70%	8.16%
75% quantile	0.30%	6.19%	9.98%	13.18%
95% quantile	8.57%	16.55%	20.50%	21.78%

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The goal of these experiments is to illustrate the superiority of the robust heuristic solution compared with stochastic solutions. Note that we have also conducted supplementary experiments to understand the expected value of additional information (EVAI) in a pooling context, which is defined as the loss incurred due to incomplete information about the distribution, under various distributions. These supplementary experiments can be found in Section EC.5 of the e-companion.

6.1. Experiments with Realistic Networks

We test our robust solution against stochastic solutions that assume a specific distribution. We conduct these tests on a network that is constructed to resemble the distribution network of a major U.S. e-commerce retailer. E-commerce retail is a good application for distributionally robust strategies since e-commerce demand distributions are often difficult to estimate as opposed to demand in physical stores. Reasons for this include the ease with which online customers could choose to purchase from any of multiple competing e-tailers, the use of dynamic price-matching strategies and flash promotions, recommendation engines that manipulate click-streams, and so on. Hence, the empirical distribution of past sales is a less reliable estimate of the distribution of future e-commerce demands. We construct the networks in our experiments based on publicly available data from Chen (2017), who use unofficial data of a U.S.-based online retailer's fulfillment center network. The data set contains information about locations of the fulfillment centers, population of the U.S. by zip code, and estimated shipping costs based on UPS Ground and UPS Next Day Air from the fulfillment centers to customer locations.

6.1.1. Model Setup. We consider networks of size $n = 10$ by choosing n random fulfillment centers from the 87 fulfillment centers available in the data. Since the results of our experiments depend on network characteristics, we take a sample of 10^2 networks and conduct our experiments for each sample, reporting the mean values over all the networks for the metrics considered.

For each network, customer locations are approximated by zip codes, and each is assigned to the nearest fulfillment center. That is, a fulfillment center can fulfill the demand realized at the assigned customer location with a per-unit cost equal to the in-location fulfillment cost s_0.

Similar to Jasin and Sinha (2015), the fulfillment costs are linear in the distance. Specifically we have $s_{ij} = s_0 + \lambda \cdot r_{ij}$, where r_{ij} is the distance in miles and $\lambda = 0.005$ is the distance sensitivity factor, with in-location fulfillment done at cost $s_0 = \$10$. This gives fulfillment costs in the range of $[\$10, \$23.6]$ for the entire network. The overage and underage cost parameters are taken to be $h = \$10$, $p = \$50$. We use hierarchical agglomerative clustering (Algorithm 1 in the e-companion) to generate the L-level nested fulfillment cost structure with $L = n$ as the base case. Because we have nested costs, we use Lemma 1 to evaluate the cost of an inventory decision $\mathbf{y}$.

The expected costs $\mathbb{E}_f[C(\mathbf{y}, \tilde{\mathbf{D}})]$ of the heuristics are calculated under the true distribution, where f is a joint distribution where all the marginal distributions have the same parametric form, and their dependencies are described by a Gaussian copula. To see the impact of the distribution on our results, we consider four cases: all marginal distributions are normal, exponential, lognormal, or gamma. The mean demands at customer locations (approximated by zip codes) are the population in millions. The coefficient of variation is taken to be equal to 1. Given mean m and variance v in a customer location, we specify the marginal distribution parameters as follows:

1. Normal: Mean $\mu = m$, variance $\sigma^2 = v$;
2. Exponential: Mean $\frac{1}{\lambda} = m = \sqrt{v}$;
3. Lognormal: Mean $\mu = \log(\sqrt{\frac{m^2}{m^2+v}})$, variance $\sigma^2 = \log(1 + \frac{v}{m^2})$;
4. Gamma: $k = \frac{m^2}{v}$, $\theta = \frac{m}{v}$.

We generate a random correlation matrix based on Numpacharoen and Atsawarungruangkit (2012) such that the maximum correlation coefficient has an absolute value less than 0.4.

6.1.2. Investigating the Approximation Error of a Nested Cost Structure. We calculate the relative gap in expected total fulfillment costs under the actual fulfillment costs and the approximate nested fulfillment costs. Given the inventory levels, if the gap is found to be small, we can confidently use the nested approximation and our heuristic in Section 5 to approximate the robust inventory levels for networks with distance-based fulfillment costs.

Indeed, we see that the nested cost structure with $L = n$ tightly approximates the expected total fulfillment cost for a variety of distributions such as normal, exponential, lognormal, and gamma (Figure 9(a)). The approximation is deteriorating as fulfillment costs become more sensitive to distance; however, even for high values of the distance sensitivity factor λ, the relative gap in expected total fulfillment costs is less than 3%. (We note that the distance sensitivity factor for UPS Ground shipping is 0.0005 as estimated by Jasin and Sinha (2015), which is ten times lower than the lowest value of distance sensitivity considered in this experiment.) Thus, nested fulfillment costs can well-approximate most distance-based shipping costs seen in practice.

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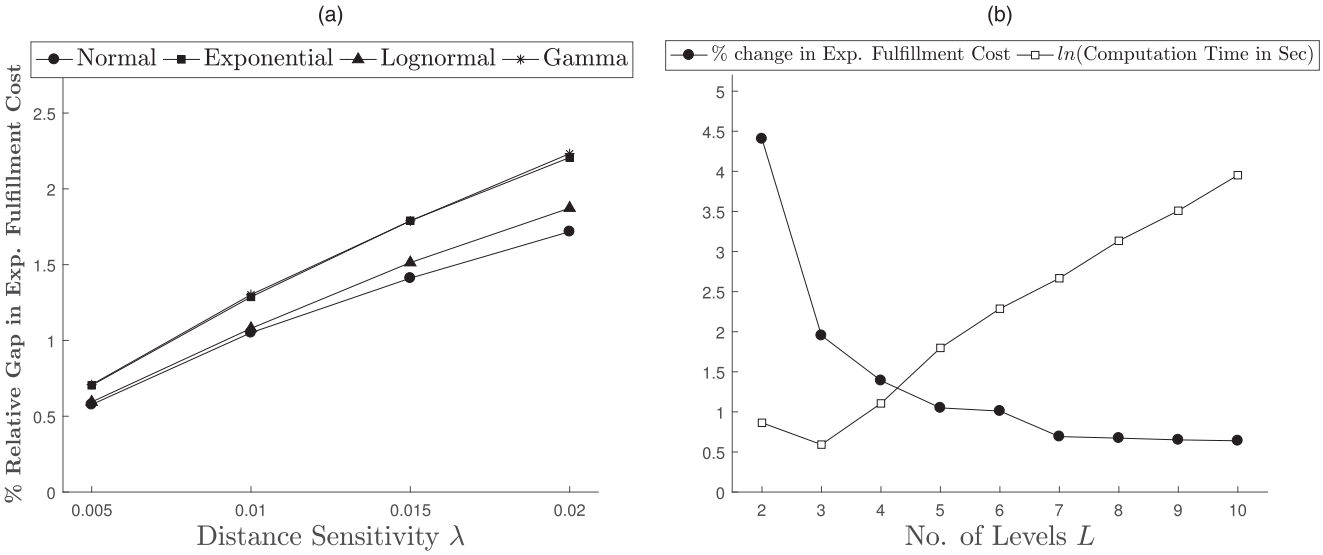
Figure 9. The Approximation of the Expected Total Fulfillment Cost by a Nested Fulfillment Cost Structure



Notes. Panel (a) shows the relative gap in expected total fulfillment cost under a variety of distributions as fulfillment costs become more sensitive to distance. Panel (b) shows the quality of approximation and computational time as the number of levels in the nested structure is varied.

Govindarajan et al. (2020, proposition 5) show that the error from assuming a constant fulfillment cost diminishes to zero as the number of locations grows while holding positive safety stock. This is because, as the number of inventory nodes in a given area increases, the chance that a unit of unfulfilled demand from one location is fulfilled from a close-by location is high. In our case, a similar intuition applies: the probability that fulfillment happens in higher levels is low, as there is enough supply to fulfill the pooled demand in lower levels. As a result, the error contribution to the expected total fulfillment cost from higher levels of fulfillment is low.

Larger values of L best approximate general fulfillment costs, although at the expense of increased computational effort in solving (3). We next test how the approximation and the computational time are affected by the number of levels. Results for an exponential distribution are shown in Figure 9(b). As expected, increasing L improves approximation quality, albeit marginally for $L > 7$. However, the computational time of our proposed heuristic in Section 5 increases exponentially. Hence, nominal values of L can achieve good approximations in relatively shorter time.

6.1.3. Misspecifying the Distribution. Knowing only the mean $\mathbf{m}$ and the covariance $\mathbf{\Sigma}$, we compare two heuristics:

1. The robust inventory solution $\hat{\mathbf{y}}$ derived from our proposed heuristic (Algorithm 2 in the e-companion);
2. The stochastic inventory solution $\mathbf{y}^{\mathrm{N}} = \operatorname{argmin}_{\mathbf{y}\geq 0}\mathbb{E}_f[C(\mathbf{y}, \tilde{\mathbf{D}})]$ that assumes f is the normal distribution (the solution is approximated using sample average approximation),

by computing their expected costs under the true distribution, which is either normal, exponential, lognormal, or gamma. We estimate the mean and covariance from 10^3 samples of the demand vector. The stochastic solutions are calculated based on a sample average approximation linear program using these training samples, and the robust solution is calculated based on the partitioned statistics estimated from the training samples. The inventory solutions are then evaluated through simulations based on 10^3 test samples generated in a similar fashion to the training samples.

The results are shown in Figure 10. Each circle corresponds to a randomly chosen network of size $n = 10$. Indeed, if the true demand distribution is normal, then $\mathbf{y}^{\mathrm{N}}$ will be the true optimal solution, in which case the relative gap in expected cost achieved by the robust solution is negative. However, this is not usually the case in reality, as the real joint distribution of demands can seldom be accurately predicted. We see that for certain networks, when the true distribution is non-normal, significant reduction in expected costs can be realized by using the robust solution instead of the normal solution. The savings are likely to be higher for larger networks as the normal distribution perceives higher pooling benefits which may not be the case under the true distribution.

Based on these numerical experiments on networks inspired from realistic settings, we can conclude that, in practice, it is better to take a distributionally robust approach when only lower order moment

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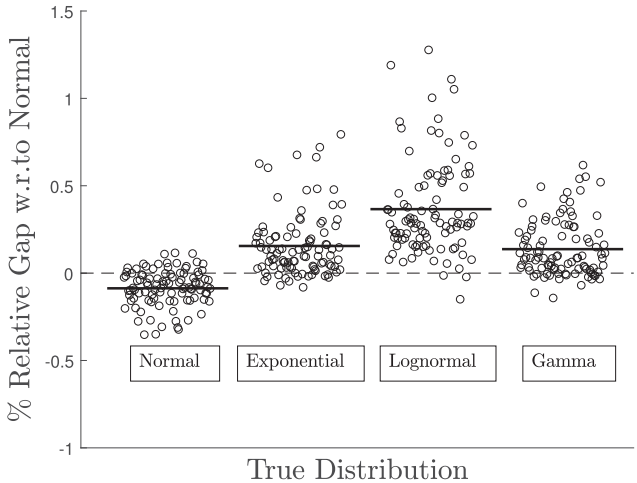
Figure 10. The Relative Gap in Expected Costs under the True Distribution of the Robust Solution $\hat{\mathbf{y}}$ and the Normal Solution $\mathbf{y}^{\mathrm{N}}$



Note. The x-axis spread of the data is solely for visual clarity.

information such as mean and covariance can be reliably estimated, rather than to assume a multivariate normal distribution, which is a commonly used approach.

6.1.4. Dynamic Myopic Fulfillment. The experiments that we conduct in Section 6.1.3 on the e-commerce fulfillment networks assume that the network flows are only decided after all the demands are realized. Although this may be true in some inventory pooling settings (e.g., a make-to-order system where $\mathbf{y}$ is the daily plant capacity), in other settings, each demand unit may arrive sequentially and the decision of how to fulfill each demand unit is made dynamically. In the latter setting, the single-period cost $\mathbb{E}_f[C(\mathbf{y}, \tilde{\mathbf{D}})]$ is a lower bound on the cost under any dynamic fulfillment policy for a given $\mathbf{y}$. This is because any dynamic fulfillment policy is deciding fulfillment on-the-fly without knowing the total demand realization $\mathbf{D}$, whereas $C(\mathbf{y}, \mathbf{D})$ defined in (2) is the smallest possible cost *after* the total demand $\mathbf{D}$ is known.

Assuming that the inventory levels at the start of the selling horizon are given by the robust solution $\hat{\mathbf{y}}$ with no replenishments allowed, we compare the expected single-period cost $\mathbb{E}_f[C(\hat{\mathbf{y}}, \tilde{\mathbf{D}})]$ to the expected cost under a dynamic fulfillment policy. If this gap is small, then the robust optimal solution of the single-period model is a reasonably good approximation to the robust optimal solution of a model with dynamic fulfillment. In these experiments, we use a dynamic *myopic* fulfillment policy which chooses the closest (i.e., lowest cost) in-stock inventory node to fulfill an arriving demand unit. We chose this specific fulfillment policy for two reasons. First, the gap against a more sophisticated dynamic fulfillment policy can only be smaller, so our experiments will give us a conservative baseline. Second, dynamic myopic fulfillment is a common practice in fulfilling e-commerce demand, especially as same-day or next-day delivery becomes more popular.

We generate a sequence of customer arrivals as follows: First, we sample a demand vector $\mathbf{D} = (D_1, \ldots, D_n)$ from the true distribution f. Note that there are a total of $\sum_{i=1}^{n} D_i$ customer arrivals. We randomize the sequence of locations in which these customers arrive by sampling a random permutation of the following set of locations:

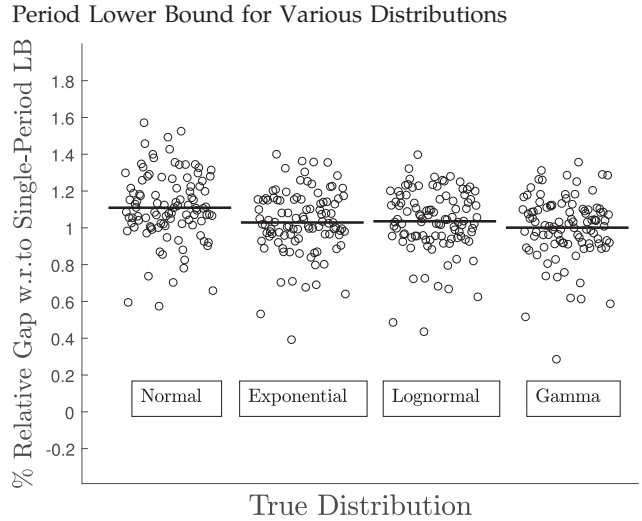
$$\left\{ \underbrace{1, \ldots, 1}_{D_1 \text{ customers}}, \underbrace{2, \ldots, 2}_{D_2 \text{ customers}}, \ldots, \underbrace{n, \ldots, n}_{D_n \text{ customers}} \right\}.$$

Once the sequence of locations is obtained in the above fashion, fulfillment decisions are taken for each unit of demand without knowledge of the future sequence of demands. Note that a myopic policy is also nonanticipatory in nature (current decisions are agnostic to future states), and a unit of demand is lost only when there is no inventory in the network that can fulfill the demand.

We use the same cost parameters and distributions as in Sections 6.1.1 and 6.1.3. We see in Figure 11 that the relative gap in expected costs between the single-period and dynamic settings is less than 2%, and hence, the single-period expected cost can serve as a good approximation for the expected cost under a dynamic setting.

We note that the myopic strategy need not be the optimal fulfillment strategy in a dynamic setting—in some cases, it might be ex post optimal to withhold a unit of inventory at a location in anticipation of future local demands that can be fulfilled at lower costs,

Figure 11. Relative Gap in Expected Costs between the Dynamic Setting under Myopic Fulfillment and the Single-Period Lower Bound for Various Distributions



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rather than costly spillover fulfillment at the current instant. As a result, the actual relative gap in expected costs will be less than 2% when the optimal fulfillment strategy is followed.

7. Conclusion

With increased focus on operational flexibilities in recent times, firms' abilities to incorporate risk pooling into their operational planning has gained importance. Given the difficulty in accurately forecasting demand distributions in various situations, a distributionally robust strategy that takes into account risk pooling is warranted, as the effect of risk pooling on operational decisions is dependent on the distribution of demands being pooled.

We provide a framework to analyze the distributionally robust multilocation problem where there are network flows after realization of uncertainty. Analyzing the two-location setting yields closed-form bounds, as well as a family of worst-case joint distributions with six support points.

For the multilocation case, we provide a heuristic approximation and upper bound for the case where the fulfillment costs exhibit a nested structure, where the cost function can be written as the sum of piecewise linear terms. We show how any general fulfillment cost structure can be approximated by this nested fulfillment cost structure through simple agglomerative clustering algorithms and that the approximation of the expected total fulfillment cost is empirically tight for commonly seen distance-based shipping cost structures under various distributions. To the best of our knowledge, this yields the first computationally tractable heuristic for the distributionally robust multilocation problem on general networks when only mean and covariance is known.

Several extensions to our work are possible. Our model and heuristic can be modified to include multiple products—the second-stage cost can be decomposed by product in the absence of capacity constraints, although the robust model does not decompose as demands are correlated. We can also extend our work to include multiple demand classes, including those that cannot be flexibly fulfilled (e.g., in-store demands that can only be fulfilled from the location at which the demand occurs). The reader is referred to Section EC.7 in the Appendix for a detailed discussion of this extension. Finally, if the moments are estimated from data, there may be uncertainty around those estimates. These can be incorporated easily into our models through linear uncertainty sets for the estimated mean and covariance matrices, similar to Natarajan et al. (2011).

Multiple directions for future work exist. Although we show that any general nested fulfillment cost structure can yield tractability to our problem, finding the optimal structure of the nested network that best approximates the general network is an open question. Another interesting future direction is the initial inventory problem under a multiperiod setting, since by writing a time-space model, this can be thought of as pooling for multiple time nodes. In Section EC.4, we demonstrate this equivalence for the two-period model where we solve for the robust initial inventory as an SDP. However, the costs in this case are asymmetric, so the n-period case cannot be simplified into a nested structure, and our heuristic does not apply. Although tractable formulations can elude us, we can approximate future stages through an affine approximation, where the future actions are restricted to be affine functions of the corresponding data (Ben-Tal et al. 2004). Under such settings, robust fulfillment decisions can be analyzed which can yield helpful tools for practitioners. Our heuristic also yields the probability of stockout at the end of the period for each node in the nested structure, which can also be used to guide dynamic fulfillment. Another natural extension is to consider how the network should look in the first place, which is the focus of process flexibility models: the solution from the inventory optimization can valuably inform network design decisions.

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